25 LOs

LEARNING OBJECTIVES

PATHOGENESIS (normal disease state)

1. Describe the structural and functional anatomy of the lower GU system of individuals assigned male at birth (including NV supply), the anatomic changes which occur in these structures across different life stages and how these changes clinically impact other abdominopelvic structures. ANAT **Connor**
2. Describe the cells and tissues of the organs of the urogenital system of individuals assigned male at birth, the changes they undergo at different life stages and identify them by microscopy. ANAT **Connor**
3. Describe the structure, cellular localization and signaling mechanisms of the androgen receptor. BCM Sophie
4. Describe the endocrine regulation of testicular function by GNRH, FSH, LH, testosterone, and inhibin. PHYSIO Sophie
5. Identify the cell of origin and biosynthesis of testosterone and its physiologic effects on target tissues. PHYSIO Sophie
6. Compare and contrast benign prostatic hyperplasia versus prostatic adenocarcinoma: location in the prostate where each tends to occur, the relationship of each to androgens, gross and microscopic morphology, and potential complications. PATH **Shannon**
7. Recall the pathophysiology of prostatitis. PATH -**Tomiwa**
8. Recall the common bacterial causes of urinary tract infection and their microbiology and virulence factors. INIM- **Tomiwa**

EPIDEMIOLOGY (incidence, prevalence, risk factors)

1. List risk factors for the development of prostate cancer and state how the presence of these factors may affect how you would counsel a patient on prostate cancer screening. CM- **Tomiwa**

CLINICAL PRESENTATION (signs and symptoms)

1. Compare and contrast the history and physical exam findings in patients with prostate conditions, including prostatitis, benign prostatic hyperplasia, and prostate cancer. CS **Shannon**
2. Describe the consequences of androgen undersecretion or deprivation in the postpubescent patient. CM- **Ayla P**

DIAGNOSIS (clinical reasoning, differential diagnosis, diagnostic tools)

1. Recall the definition and application of the following features of diagnostic testing characteristics in the context of screening for prostate cancer: sensitivity, specificity, positive predictive value, negative predictive value. EBM **Ravi**
2. Identify the structures in the abdominopelvic region via imaging technologies. ANAT - **Ayla P**

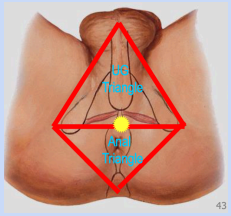
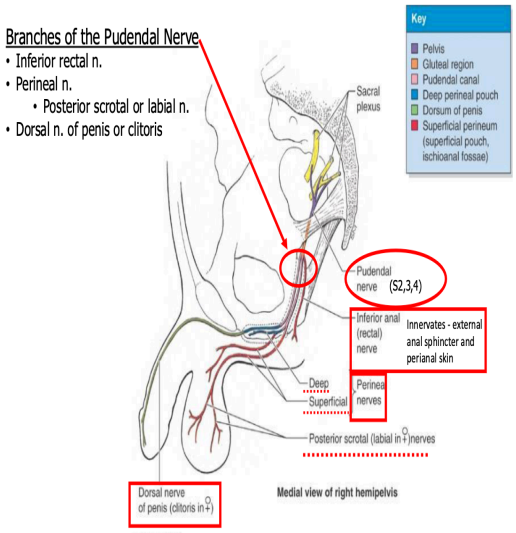
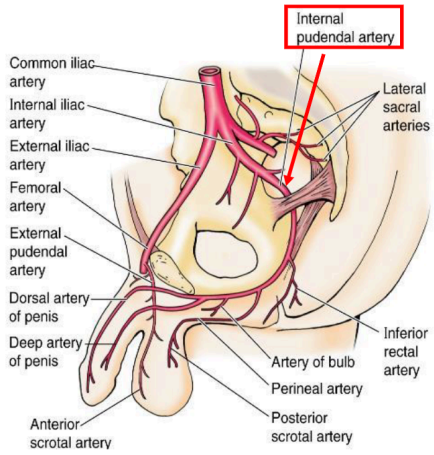
3. Compare common etiologies for hematuria in different sexes and describe the diagnostic approach. CM - **Spencer**
4. Discuss the differences between histopathologically detected prostate cancer and clinically significant prostate cancer, and explain how grade group, tumor volume, and other pathologic features determine which patients are appropriate candidates for active surveillance versus definitive treatment. PATH and CM **Shannon**
5. Describe the Gleason system of grading prostate cancer, including the histopathological diagnostic criteria for the diagnosis of prostatic adenocarcinoma, and how it relates to tumor biology. PATH **Shannon**

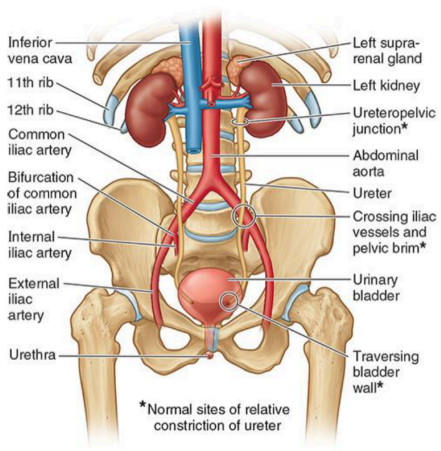
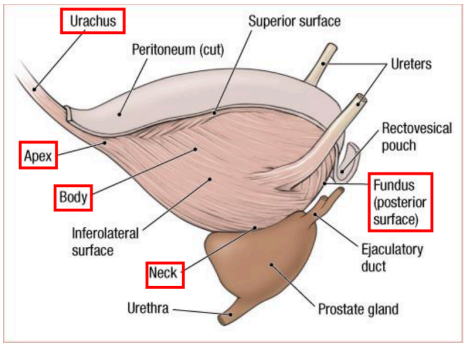
TREATMENT and OUTCOMES

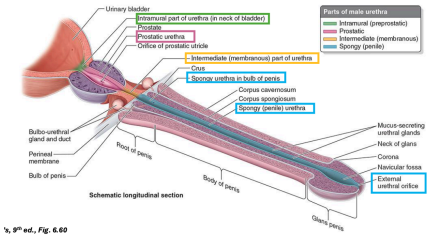
1. Compare and contrast the MOA and side effects of alpha blockers and 5-alpha reductase inhibitors. CM and PHARM **Ishaan**
2. Describe the principle of active surveillance for prostate cancer. CM **Ravi**
3. Describe the utility of PSA for surveillance/monitoring in patients who underwent or are undergoing prostate cancer treatment. CM **Ravi**
4. Demonstrate an understanding of the obligation of error disclosure (see reading in additional resource folder). PROF JP Done
5. Review the classification of medical errors and the epidemiology of common errors. HSS. (Adverse Events, Near Misses, and Errors by the Agency for Healthcare Research and Quality, available at: https://psnet.ahrq.gov/primer/adverse-events-near-misses-and-errors) JP Done
6. Describe the elements of full disclosure and apology when addressing those affected by medical errors. HSS. Reading: Chapter 18: The Malpractice System in Wachter, R. M., & Gupta, K. (2018). Understanding patient safety. **Only focus on section: Error Disclosure, Apologies, and Malpractice.** Available as e-text, Scott Memorial Library. Link to full book (please navigate to chapter/section mentioned above): JP Done

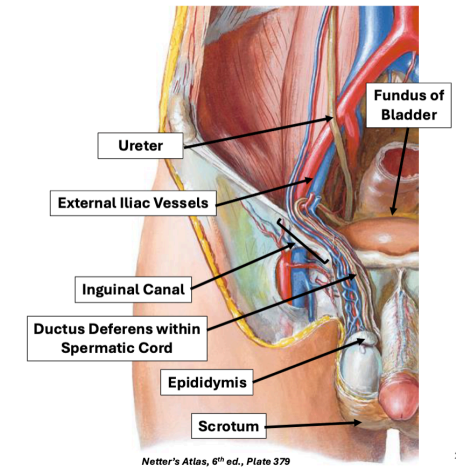
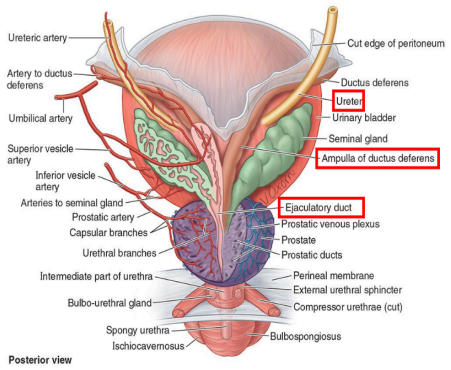
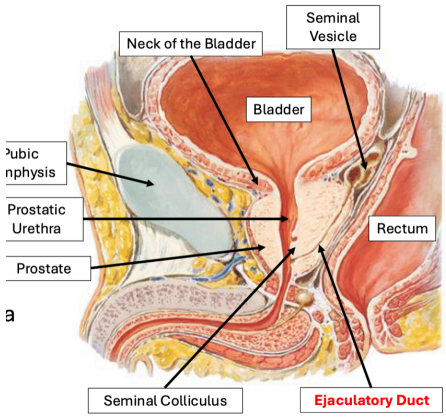
https://jefferson.primo.exlibrisgroup.com/permalink/01TJU_INST/1c1edf2/alma991000796882403866
7. Describe the quality-of-life implications of prostate cancer treatment for the different treatment modalities. CM - **Spencer**
8. Explain the mechanisms of action, adverse effects (major and common), and rationale for combined use of drug classes used as androgen deprivation therapy (chemical castration). PHARM **Ishaan**
9. Describe the mechanism of action, adverse effects, and clinically important drug interactions of PDE5 (phosphodiesterase 5) inhibitors in the treatment of erectile dysfunction. PHARM **Ishaan**

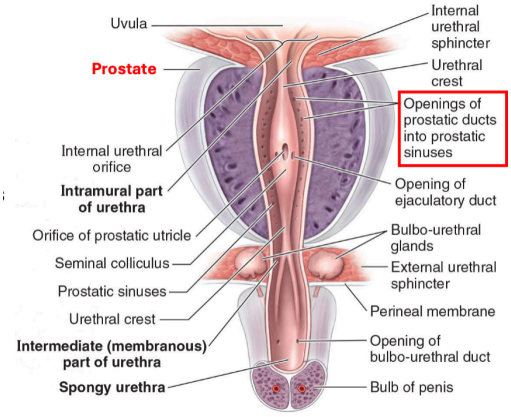
Describe the structural and functional anatomy of the lower GU system of individuals assigned male at birth (including NV supply), the anatomic changes which occur in these structures across different life stages and how these changes clinically impact other abdominopelvic structures. ANAT **Connor**

Structure	Blood supply	Innervation	
Perineal triangle - external content: scrotum & penis 	Branches of internal pudendal artery: • Inferior rectal • Perineal: posterior scrotal or labial • Artery of bulb of penis • Deep artery of penis • Dorsal artery of penis	Branches of pudendal nerve: • Inferior rectal nerve → innervate external anal sphincter & perianal skin • Perineal nerve → deep & superficial branches. Superficial branch becomes posterior scrotal nerves • Dorsal nerve of penis	 

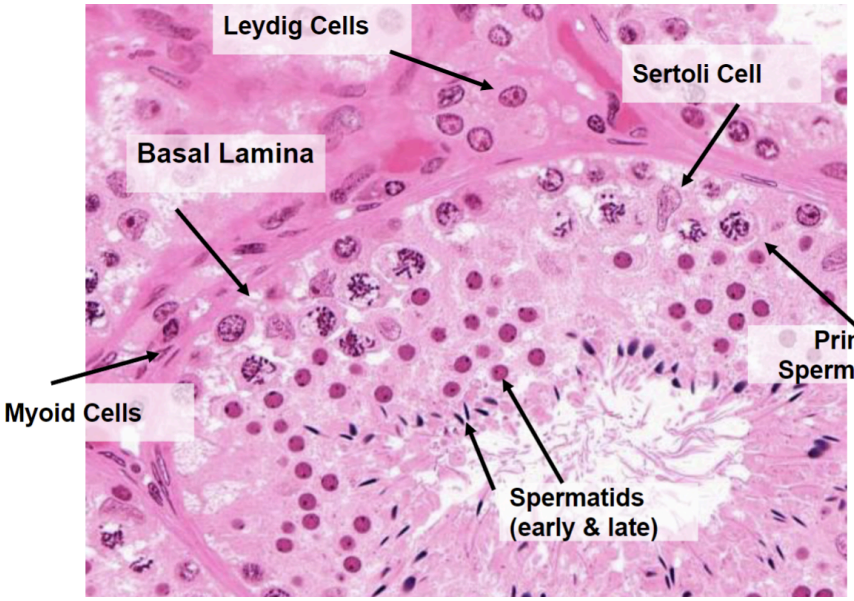
Ureter: • Muscular tubes connecting the kidney to bladder • Has 3 areas that are prone to constriction: ureteropelvic, pelvic inlet, vesicoureteral junctions	• Upper third: ipsilateral renal artery • Middle third: gonadal artery • Distal third: superior vesical artery	Sympathetic: • The lesser and least thoracic and the splanchnic nerves (T12 - L2) → aorticorenal, superior, and inferior hypogastric autonomic plexuses → ureter Parasympathetic: • Sacral pelvic splanchnic nerves (S2-S4) → synapse on target organs	
Bladder: • Muscular, hollow organ serves as a temporary reservoir for urine • When empty, it is contained in the lesser pelvis • When full, it extends into the greater pelvis • Anterior to rectum	Branches of internal iliac (anterior division) • Superior portion: superior vesical artery • Inferolateral surface, fundus, neck of bladder: inferior vesical artery and branches of vaginal artery	Sympathetic: presynaptic fibers (T11 - L2/3) travels through lumbar splanchnic nerve → prevertebral ganglia (aortic plexus) → internal urethral sphincter. **Motor innervation** (parasympathetic): presynaptic fibers (S2-4) travels through pelvic splanchnic nerve → synapse wall of bladder → contract detrusor muscle and relax internal urethral sphincter Pain innervation: Roof of bladder: sympathetic Body, inferolateral surfaces, fundus, neck, apex: parasympathetic	

Urethra: Intramural urethra: • Urine enters urethra at the internal urethral orifice • Can vary in size Prostatic part: • Located within prostate • Urethral crest • Opening of ejaculatory duct • Prostatic sinuses Membranous urethra: • Starts at the end of prostate and ends at bulb of penis • Surrounded by external urethral sphincter • Bulbourethral glands on lateral sides Spongy (penile) urethra: • Runs within corpus spongiosum • Bulbourethral gland's ducts enter the spongy urethra to provide lubrication to the area	• Inferior vesical artery • Bulbourethral artery • Internal pudendal artery	• Sympathetic: T11 - L2/3 → lumbar splanchnic fibers → aortic plexus → internal urethral sphincter, prostate, & seminal glands • Parasympathetic: presynaptic fibers (L2-4) travels through pelvic splanchnic nerves → target organs	 's, 9th ed., Fig. 6.60

Ductus Deferens: • Ductus deferens is a continuation of epididymis → travels along the spermatic cord → goes through the inguinal canal → enters deep inguinal ring → crosses external iliac vessels → pass superior to ureter → joins with seminal vesicle to form ejeculary duct → empties into prostatic urethra		Similar to above	 Netter's Atlas, 6th ed., Plate 379  Posterior view 

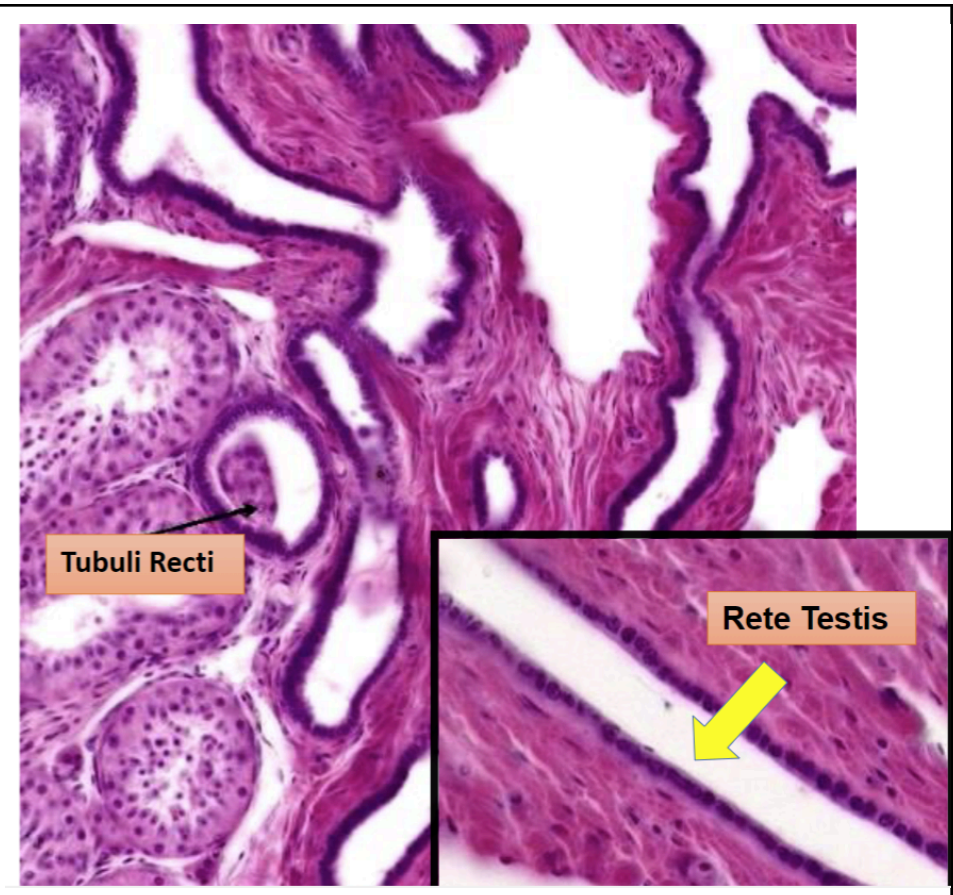
Prostate: • Largest accessory gland within testosterone-driven reproductive system • ⅔ glandular and ⅓ fibromuscular • Fibrous capsule that contains plexus of veins and nerves • Provides 20% volume of semen → transports & activates sperms	**Arterial supply**: branches from the superior and inferior vesical arteries **Venous drainage**: • Parallels the course and name of the arteries • Prostatic venous plexus drains into adjacent plexuses (vesicle) or inferior vesical veins • Ductus deferens: drains through the distal part of the pampiniform plexus → prostatic plexus	Similar to above	

Describe the cells and tissues of the organs of the urogenital system of individuals assigned male at birth, the changes they undergo at different life stages and identify them by microscopy. ANAT **Connor**

Structure	Image
Seminiferous Tubules ● Highly cellular tubules where sperm are generated ● Populated by ○ germ cells (peripheral) ○ **sertoli cells** (perpendicular to the wall of the tubule) ○ Spermatocytes (round nuclei closer to center) ○ Spermatids (small thin nuclei) ○ Myoid (surround periphery) ○ Leydig cells in interstitium ● Increase in size in puberty due to sperm production	

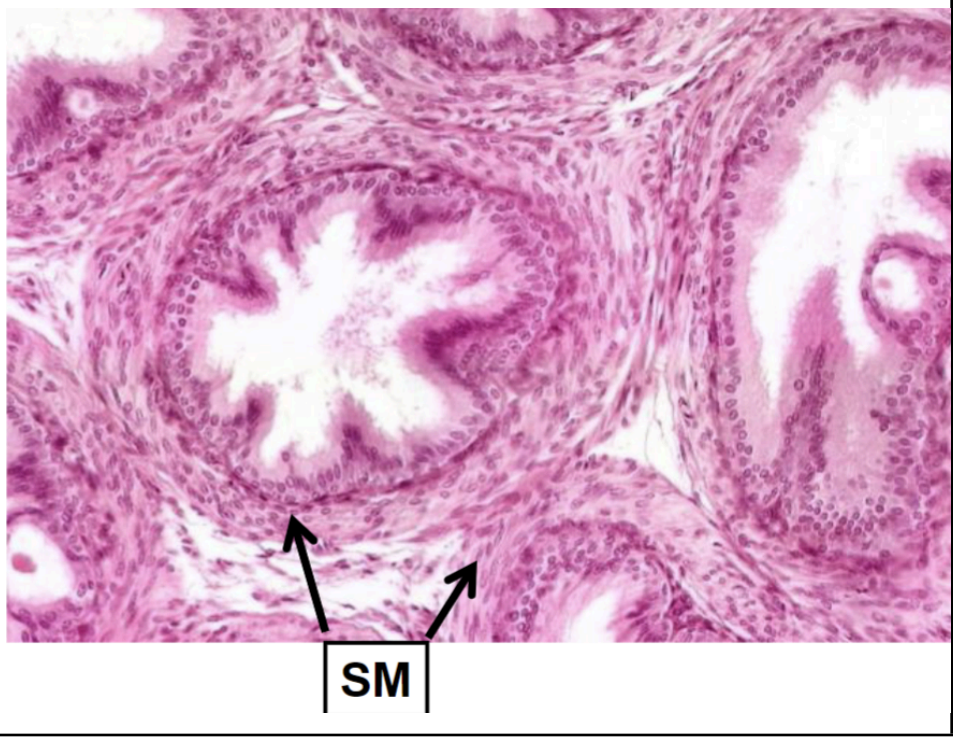
Intratesticular Ducts

- Composed of three major components
- Strait tubules
  - “tubuli recti”
  - connect seminiferous tubules to the rete testis
  - simple cuboidal epithelium
- Rete Testis
  - Carry sperm further out of the testis
  - cuboidal epithelium
- Veins
  - carry blood
  - simple squamous epithelium
- Arteries are also present - not shown here



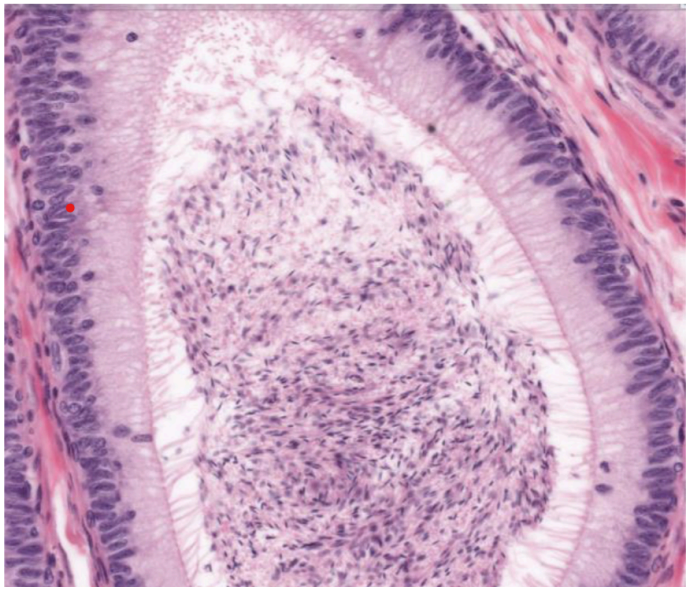
Efferent Ductule

- Connect rete testis to the epididymis
- pseudostratified columnar epithelium
- “scalloped” epithelium
  - alternating tall and short columnar cells
- surrounded by smooth muscle



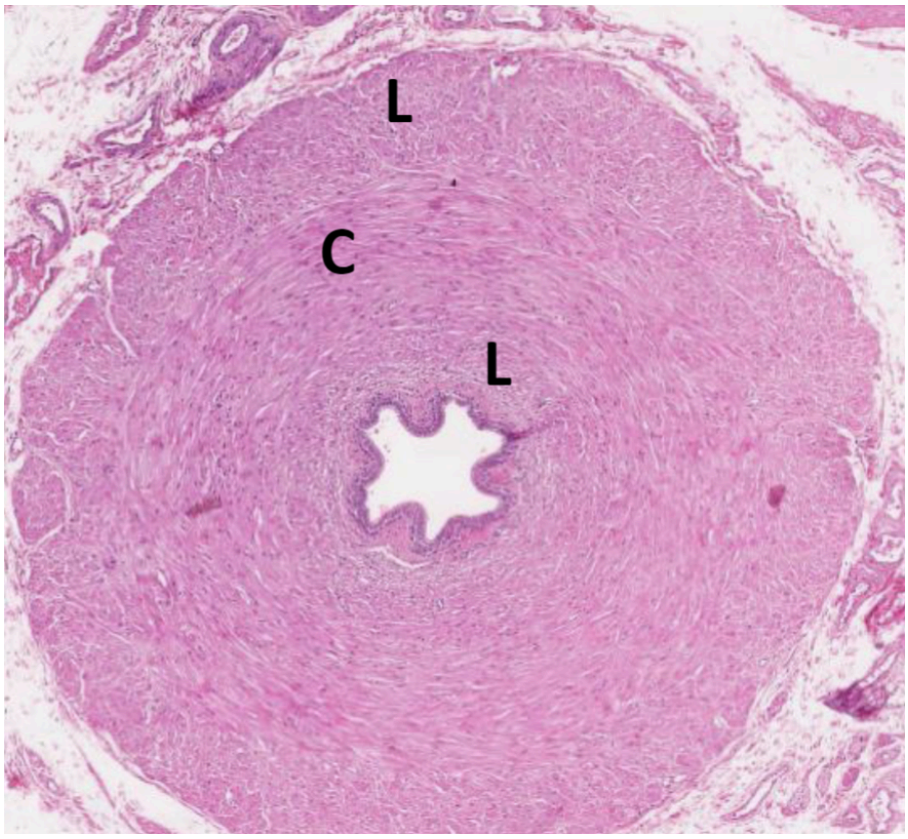
Epididymis

- Pseudostratified columnar epithelium
    - Very uniform in size compared to efferent ductules
- Stereocilia/microvilli on epithelium (for absorption of seminiferous tubule fluid)
- may be filled with sperm
- site of final sperm maturation



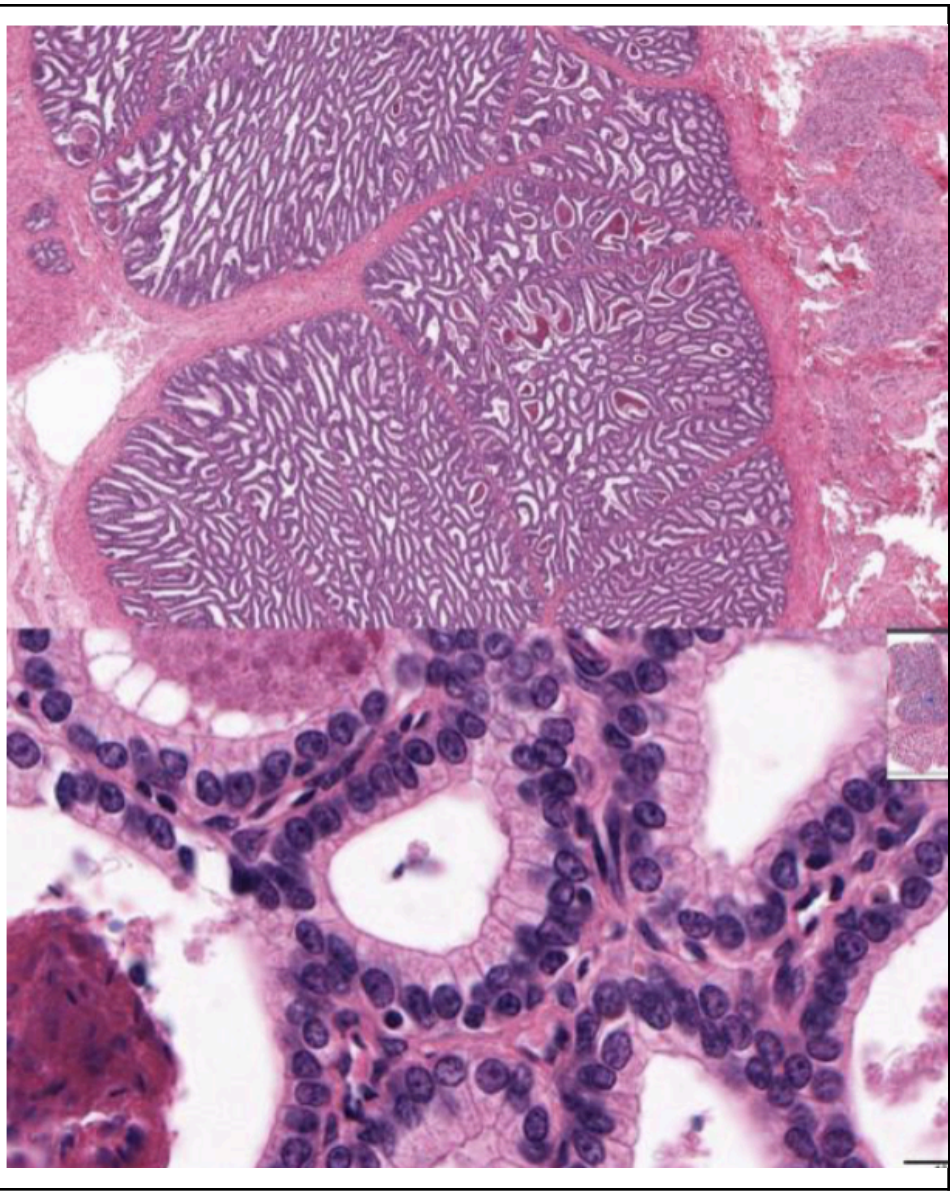
Vas Deferens

- More pseudostratified columnar epithelium
- Three layers of smooth muscle
- Longitudinal > circular > longitudinal
- connect epididymis to the urethra
- Also has an outer adventitia



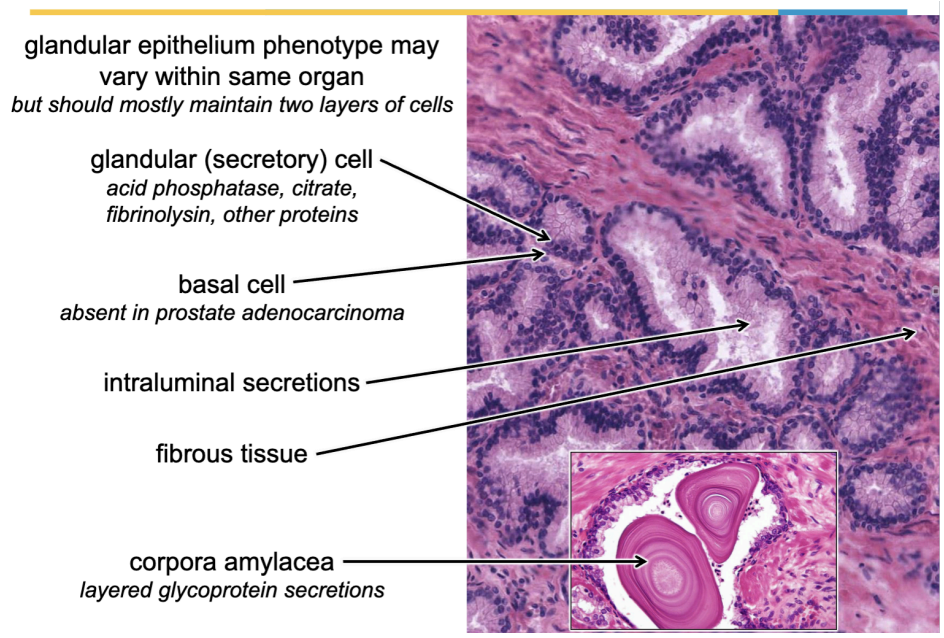
Seminal Vesicles

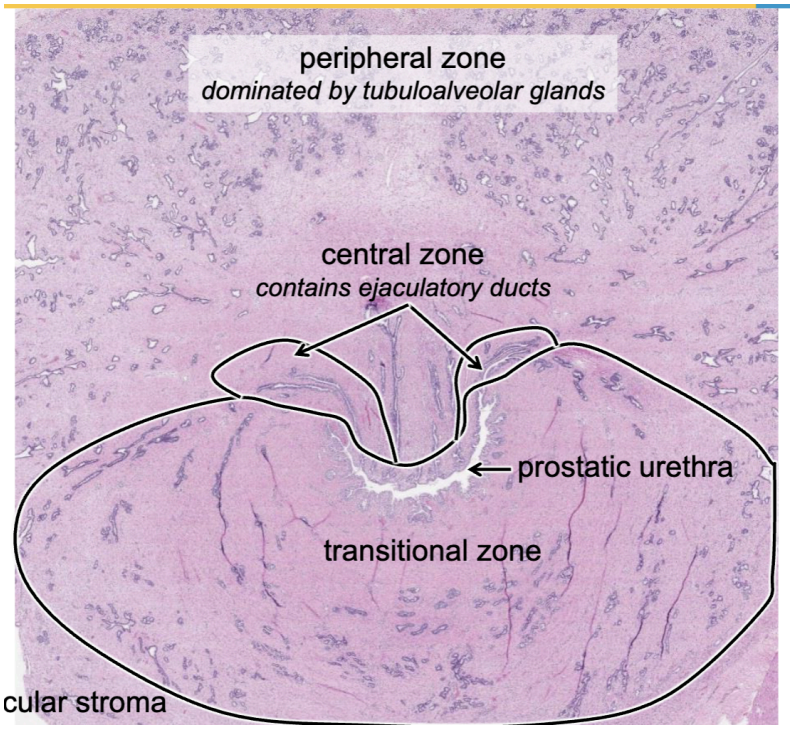
- Secrete nutrient-rich fluid for sperm
- highly folded/convoluted structure to maximize surface area for fluid secretion
- Pseudostratified columnar
    - Basal round cells are regenerative
    - tall columnar cells are secretory
- Thick, fibromuscular capsule contracts during ejaculation to release fluid



Prostate Gland

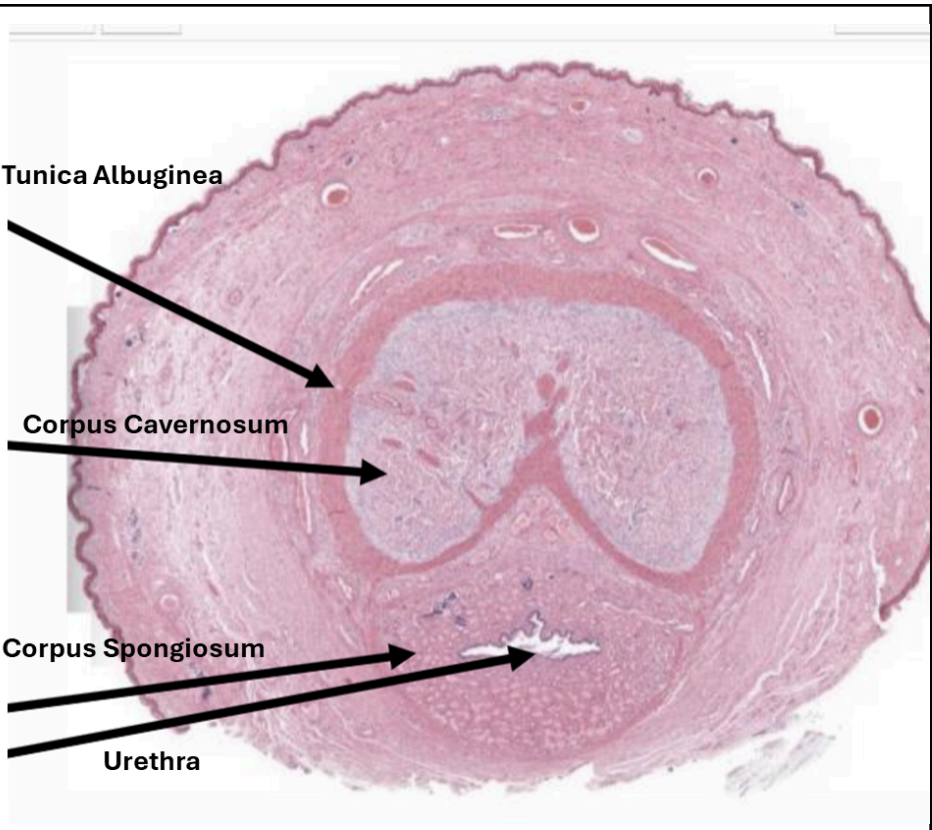
- Wide range of morphology
  - Squamous epithelium
  - pseudostratified columnar epithelium
- Glandular lumens may contain **concretions**
  - Calcified precipitate of the secreted fluid
- Fibromuscular Stroma





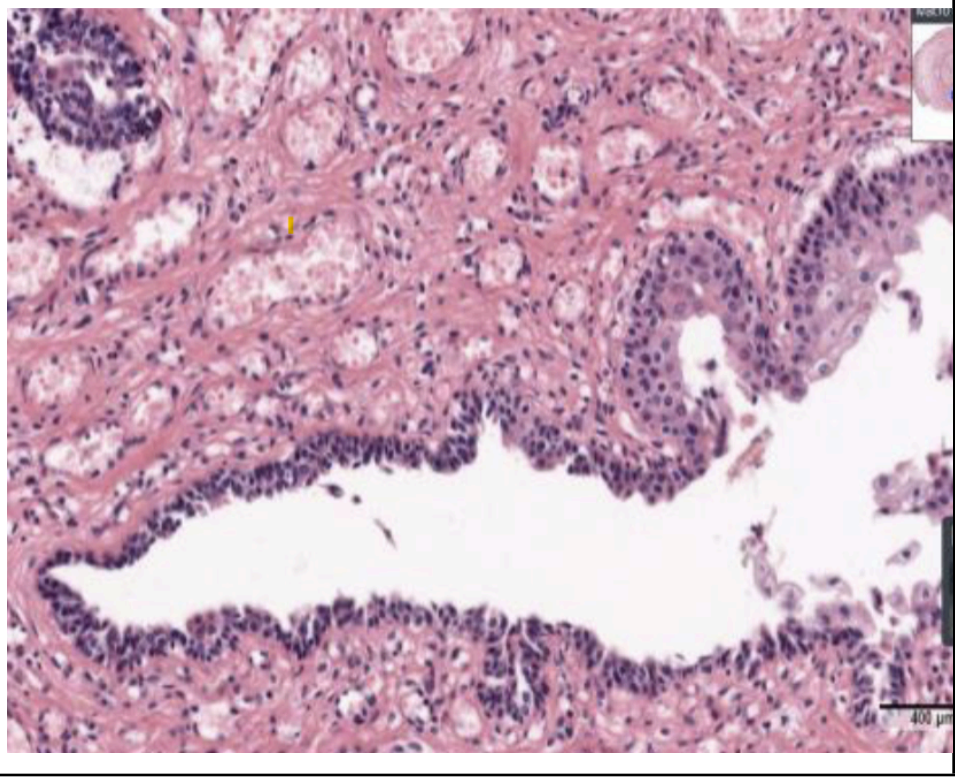
Penis

- Corpus Cavernosum
  - spongy tissue that fills with blood
  - Small helicine arteries, larger central arteries, venous sinusoids
  - Surrounded by fibrovascular tunica albuginea
- Corpus Spongiosum
  - more spongy tissue that fills with blood
  - highly vascular
  - contains urethra



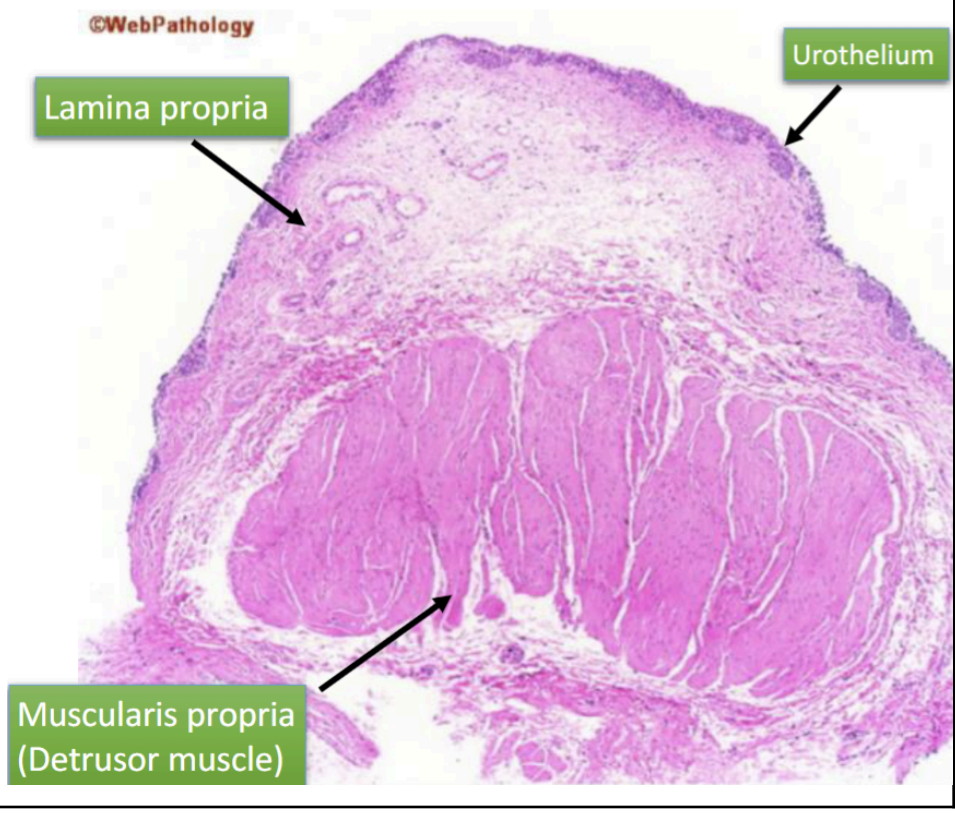
Urethra

- Carries urine from bladder to outside
- proximal urethra has transitional epithelium
- distal urethra has stratified squamous (non-keratinized)



Bladder

- Lined by urothelial (transitional) epithelium
- lamina propria (loose areolar connective tissue)
    - under epithelial layer
- Muscularis mucosa (thin smooth muscle layer under the lamina propria)
- Muscularis propria - this is the detrusor muscle. Surrounds the entire bladder



3. Describe the structure, cellular localization and signaling mechanisms of the androgen receptor. BCM Sophie

Androgen receptor

- Intracellular (ligands are steroid derivatives so lipophilic)
- Primary ligands are DHT and testosterone
    - DHT binds much more readily than testosterone
- Encoded by gene on x chromosome
- **Type 1 Nuclear Hormone receptor**
    - Homodimerizes when activated
    - Recruits co-activators

Regions:

- **N-terminal domain**: contains activating-factor-1 that is important for transcriptional activation
- **DNA binding domain**: has 2 zinc-finger domains that recognize Androgen response elements (AREs) on DNA
- **Hinge region:** flexible liker that helps transport into nucleus
- **Ligand binding domain:** binds testosterone and DHT, contains activating-factor-2, important for activating co-regulators, bound to chaperone in absence of androgen

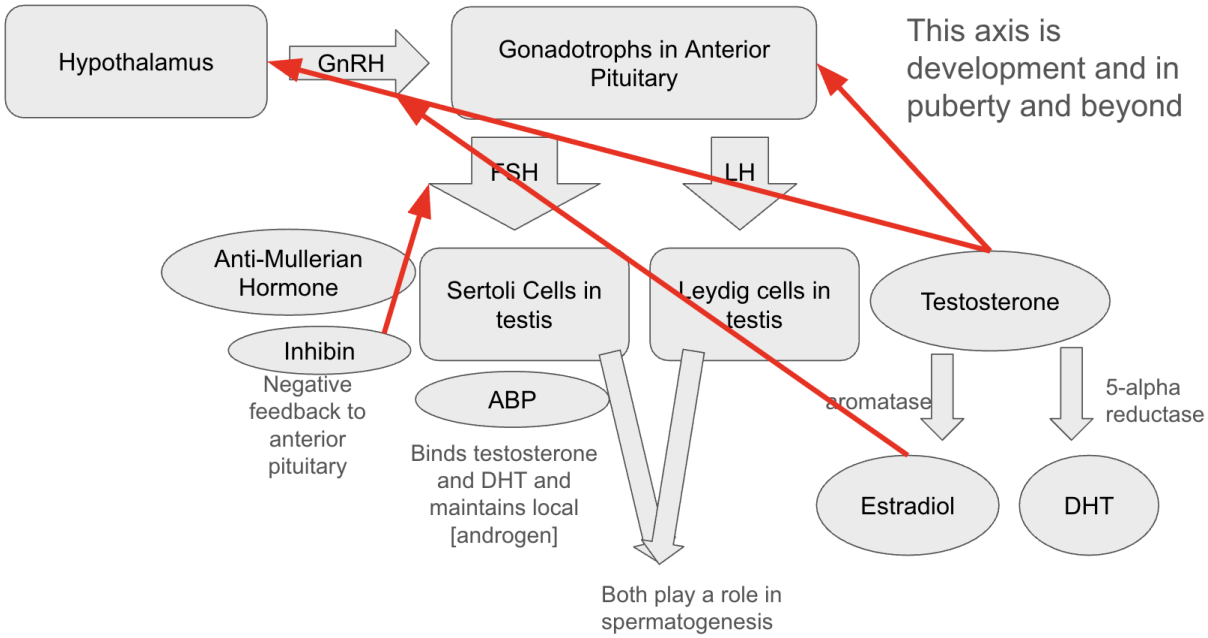
<u>In the absence of androgens</u>

- AR is in the cytoplasm
- Associates with heat shock proteins (especially HSP-90) which helps it remain available to bind ligands
- Some ARs are also in nucleus but less

<u>In the presence of androgens</u>

- **Slow Genomic signaling**
    1. Androgens diffuse across plasma membrane
    2. Androgens bind to ligand binding domain of AR, and AR undergoes a conformational change
    3. Chaperone associations are lost
    4. Goes to nucleus
    5. AR dimerizes (**homodimer)**
    6. DNA binding domain of AR binds DNA and regulates transcription
- **Rapid Non-genomic signaling**
    - Binding to AR receptor also induces more rapid responses implicating the
        - MAPK/ERK: cell differentiation, proliferation, survival
        - PI3K/AKT: metabolism, cell survival and growth
        - Intracellular calcium:
        - Interaction with signaling proteins like Src
            - Src is a co-activator to modify chromatin, facilitate transcription

4. Describe the endocrine regulation of testicular function by GNRH, FSH, LH, testosterone, and inhibin. PHYSIO Sophie



The HPG axis is PULSATILE, with approximately one pulse happening an hour

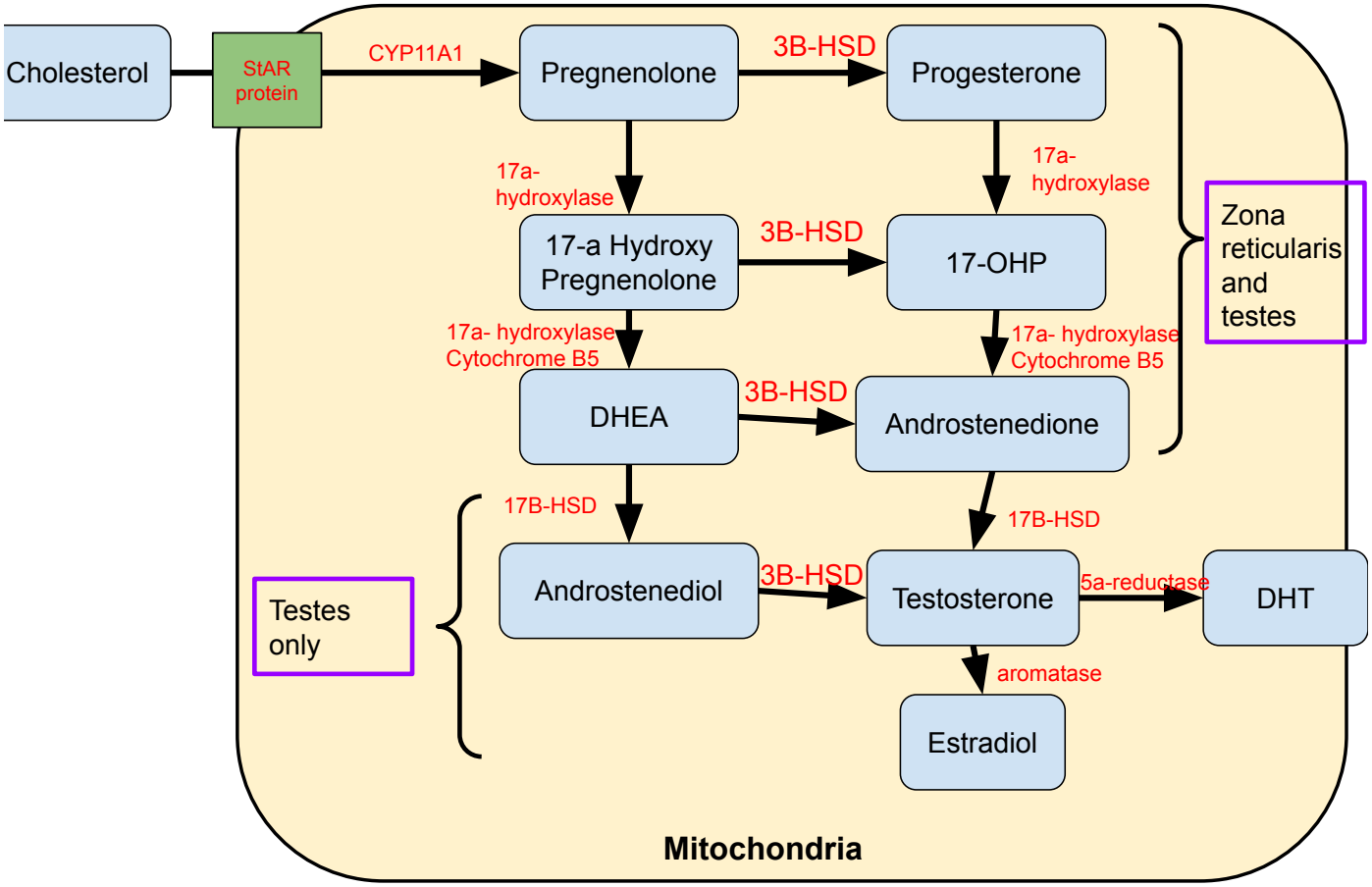
1. When the hypothalamus pulses, GnRH is released
2. **GnRH** acts on the anterior pituitary, gonadotrophs release FSH and LH
3. **FSH** acts on sertoli cells in testes
    a. Sertoli cells release
        i. **AMH**: degrades Mullerian Ducts (prevention of female anatomy from developing)
        ii. **ABP**: binds testosterone and DHT and maintains local concentrations of androgens
        iii. **Inhibin B**: provides negative feedback to the anterior pituitary
            1. Specifically inhibiting FSH
4. **LH** acts on leydig cells in testes
    a. Leydig cells release **testosterone**
        i. Testosterone is converted to **estradiol** via **aromatase**
            1. Estradiol stimulates GH and helps growth pre-puberty
            2. Estradiol also helps epiphyseal plate closure after puberty, limiting growth
            3. Estradiol also increases bone density, limiting resorption
        ii. Testosterone is converted to **DHT** via **5-alpha-reductase**
            1. DHT helps with the masculinization of external genitalia
            2. DHT promotes facial and body hair, but may contribute to head hair loss
            3. DHT activates sebum production (contributes to acne)

iii. Testosterone provides negative feedback to both the hypothalamus and anterior pituitary
iv. Testosterone provides paracrine positive feed back to the sertoli cells
    1. Drives spermatogenesis
v. Testosterone induces
    1. larynx growth and voice deepening
    2. increased muscle mass
    3. development of internal XY genitalia via preservation of the wolffian ducts
    4. increased libido
    5. increased erythropoiesis
    6. Growth of penis
    7. Inhibits breast development

5. Identify the cell of origin and biosynthesis of testosterone and its physiologic effects on target tissues. PHYSIO Sophie

Testosterone is mainly secreted from leydig cells in response to LH (see path 4 for more details on hormonal regulation of testosterone production)

Biosynthesis: *this is ugly i apologize in advance*



Testosterone effect:

- Development in utero
  - Preserves Wolffian duct
    - Develops into internal male genitalia
      - Vas deferens, seminal vesicles, and epididymis
  - Via DHT promotes development of external genitalia
- Puberty and beyond
  - **Larynx**: enlarges and deepens voice
  - **Muscles**: increases strength and mass
  - **Bones**: increased growth and signaling
  - **Penis**: increased growth
  - **Brain**: increased libido
  - **Bone marrow**: increased erythropoiesis

        - **Sertoli cells:** supports spermatogenesis
        - Inhibition of breast development
- Feedback effects:
    - Negative feedback
        - Hypothalamus
        - Pituitary
    - Positive feedback
        - Sertoli cells for spermatogenesis
    - Converted into DHT and estradiol

Compare and contrast benign prostatic hyperplasia versus prostatic adenocarcinoma: location in the prostate where each tends to occur, the relationship of each to androgens, gross and microscopic morphology, and potential complications. PATH

1. Location in the prostate

	BPH	Prostatic adenocarcinoma
Zone	**Transition zone** and periurethral tissue, including the middle and lateral lobes	**Peripheral zone**, especially posterior and posterolateral
Clinical consequence	Nodules sit around the urethra, so they compress it. This causes **obstructive urinary symptoms early**.	Tumor is far from the urethra, so it is often **asymptomatic early**. Because it is posterior, it can be **palpated on DRE** as a firm or hard area.
Focality	Multinodular	**Usually multifocal**

- An enlarging BPH nodule in the transition zone can **compress the surrounding peripheral zone and cause secondary atrophy** there
- BPH is a benign, DHT-driven hyperplasia of glands and stroma in the transition zone that retains its basal cells and causes urethral obstruction.
- Adenocarcinoma is an androgen-sensitive, infiltrative tumor of the peripheral zone with a single cell layer and no basal cells, which invades nerves, the seminal vesicles, and bone.

2. Relationship to androgens

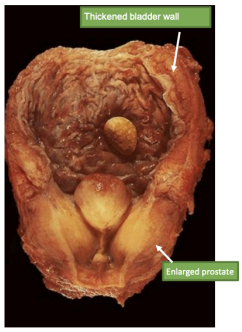
- **Both depend on androgens. Only cancer is malignant, and BPH is not premalignant.**
- **BPH** is androgen (DHT)-dependent:
    - Testosterone is converted to **DHT** by **5α-reductase type 2**, which sits in the **stromal cells**. Type 1 is found in skin and liver.
    - DHT binds androgen receptors in both glandular and stromal cells. It induces growth factors that **increase stromal proliferation and decrease epithelial cell death**.
    - BPH is rare in men castrated before puberty.
    - Therapeutic correlate:
        - **5α-reductase inhibitors** (finasteride, dutasteride) shrink the gland.
        - **α-blockers** relieve symptoms by relaxing prostatic smooth muscle; they do not change androgen signaling.
        - Some, but not all, patients respond to these drugs.
- **Adenocarcinoma** is androgen-**sensitive**, and the role of androgens is **"permissive"**:
    - Men castrated before puberty do not develop prostate cancer.
    - Androgens allow the tumor to grow but do not by themselves cause the malignant transformation.
    - The tumor responds **for a time** to surgical or chemical castration (**ADT**):
        - Bilateral orchiectomy, or
        - A **GnRH agonist** such as leuprolide. Because the agonist causes an initial testosterone surge, it is paired with an AR antagonist such as bicalutamide.
    - Eventually the tumor becomes **castration-resistant (CRPC)**.
        - CRPC shows up as a rising PSA and progression on imaging despite castrate testosterone levels, and it carries a poor prognosis.

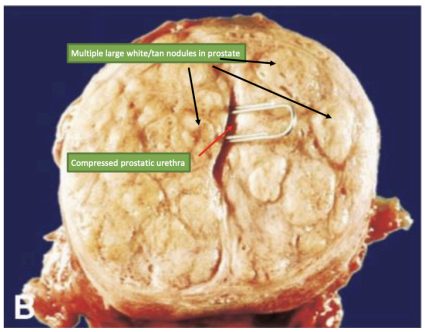
- Finasteride and dutasteride lower PSA by about 50%. You must account for this when you interpret PSA in a man being treated for BPH (Shah).

3. Gross morphology

BPH:

- Enlarged prostate with **multiple large, well-circumscribed, white/tan nodules**.
- Some nodules are solid and some are spongy or cystic.
- **Compressed, slit-like, irregular prostatic urethra**, sometimes with a prominent middle lobe protruding into it.
- **Thickened bladder wall**, which reflects outlet obstruction.
- On DRE the gland is **smooth, symmetric, and rubbery**.





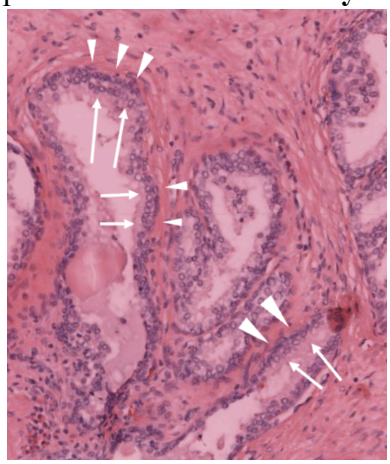
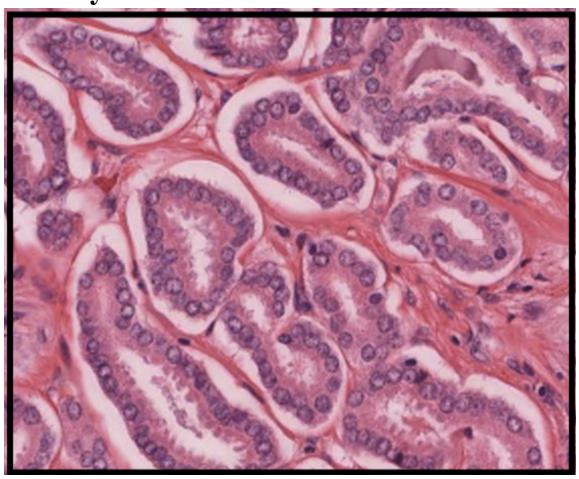
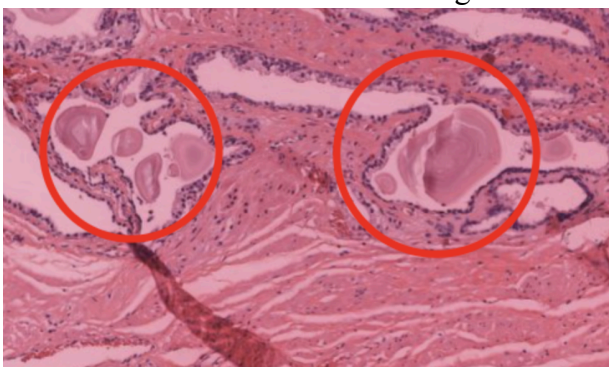
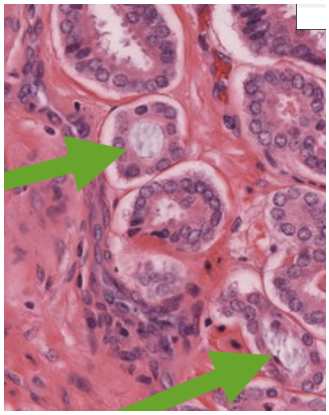
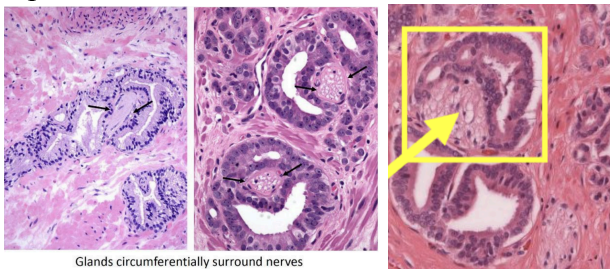
Adenocarcinoma :

- **Firm, gritty, poorly circumscribed, homogeneous solid area**, yellow-white or gray.
- Located in the **posterolateral peripheral zone**.
- Often subtle grossly; it is frequently easier to feel than to see.
- On DRE it is a **hard, irregular nodule**.

4. Microscopic morphology

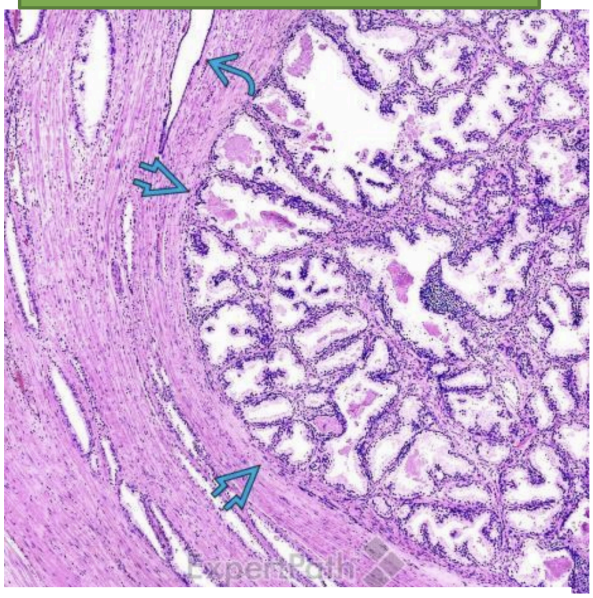
- The single most important discriminator is the **basal cell layer**: it is retained in BPH and lost in cancer.

Feature	BPH	Adenocarcinoma
Process	**Hyperplasia** of both glands and stroma. Nodules can be glandular, stromal (fibromuscular/spindle cell with many capillaries), or mixed.	**Malignant gland proliferation** that infiltrates the stroma; crifiform glands
Gland size and architecture	**Large** glands, often cystically dilated, with **papillary infoldings** and an undulating, **serrated luminal border**. Lobular architecture is preserved.	**Small, crowded, back-to-back** glands with little intervening stroma. They are haphazard and infiltrative and lack the lobular arrangement.

Cell layers	**Two layers:** inner secretory columnar cells plus an **outer basal cell layer** 	**One layer.** The basal cells are absent. 
Nuclei	Bland and small, with inapparent nucleoli	**Enlarged nuclei with prominent nucleoli**
Luminal contents	**Corpora amylacea**: glassy, laminated concretions that are a normal finding 	Can contain **blue mucin** and pink crystalloids 
Invasion	None	**Perineural invasion** is common. It is not an independent prognostic factor, but extensive perineural invasion suggests spread outside the capsule.  Glands circumferentially surround nerves

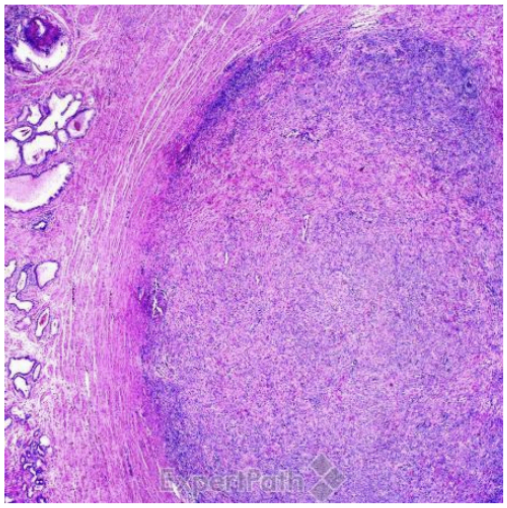
Histo

Nodule containing hyperplastic prostatic glands

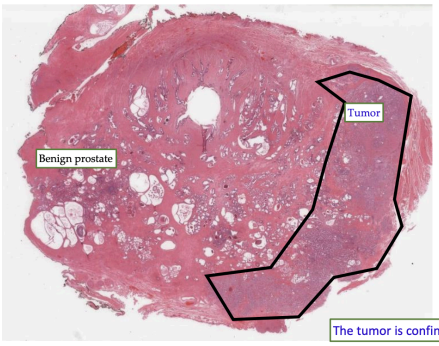


BPH nodule arising from transition zone (cyan open arrow) shows compression with secondary atrophy of the surrounding peripheral zone (cyan curved arrow). The hyperplastic glands are much larger, showing luminal undulations and papillary infoldings, and the hyperplastic stroma is denser compared with peripheral zone.

Nodule of stromal hyperplasia adjacent to hyperplastic glands

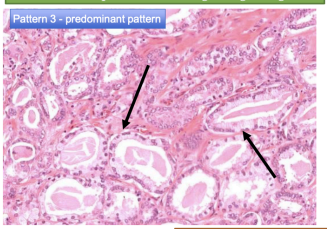


Well-circumscribed stromal hyperplastic nodule shows sheets of spindle cells containing abundant capillaries dispersed throughout the stroma.

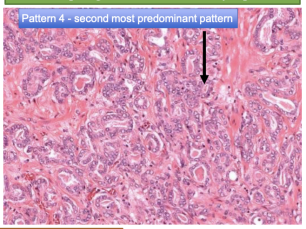


The tumor is confined within the prostatic capsule

Well-formed separate infiltrating malignant glands



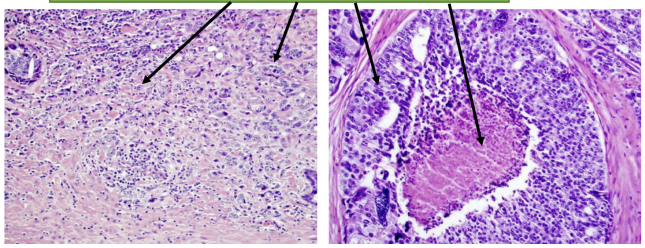
Fused, complex glands that cannot easily separated

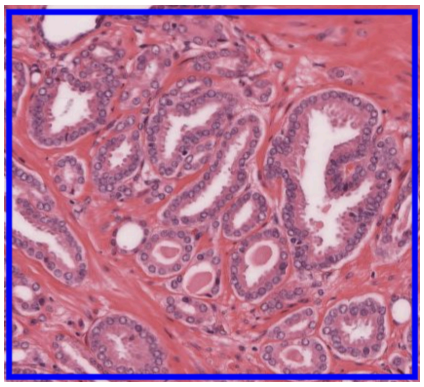


Gleason score 3 + 4 = 7 (intermediate grade)

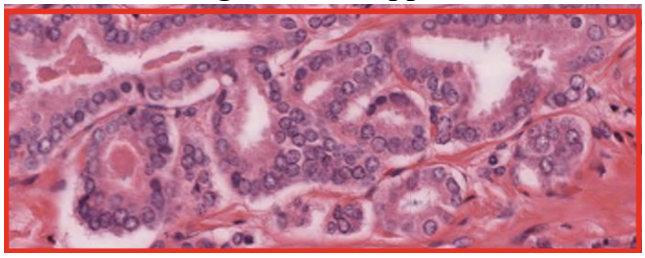
Pattern 5 – not seen in *this* patient's case

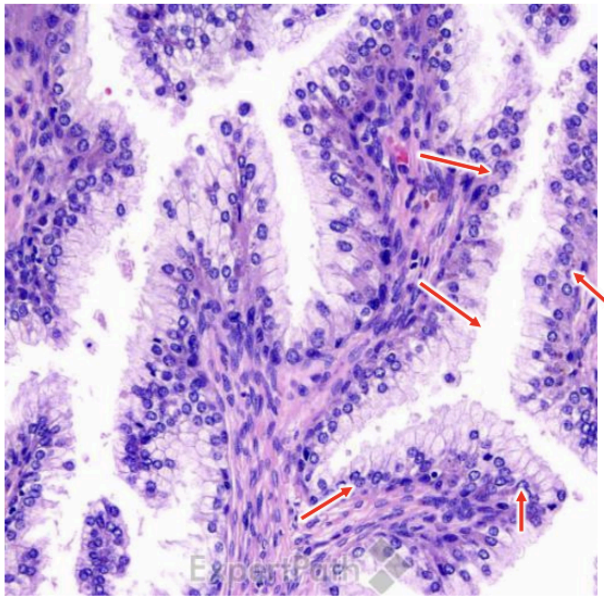
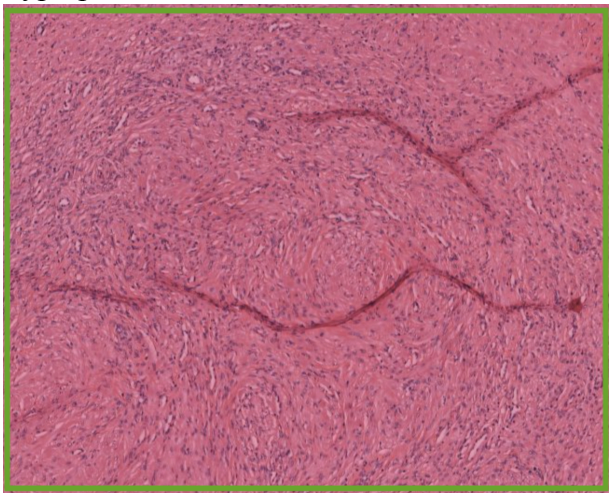
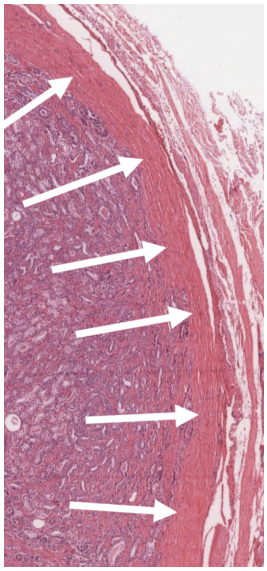
No glandular differentiation: single cells, cords, solid growth, comedonecrosis

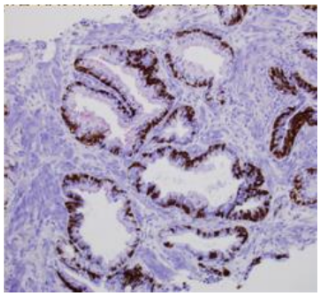
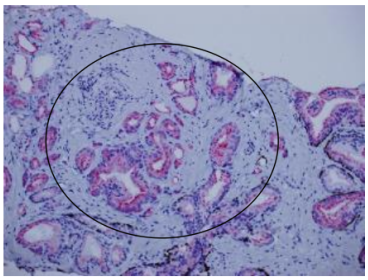
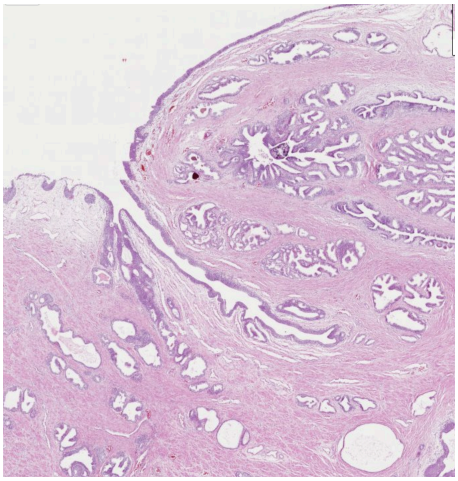
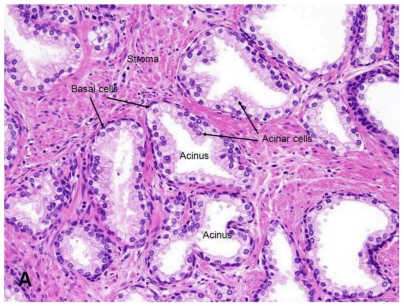




Red box showing sieve like appearance:

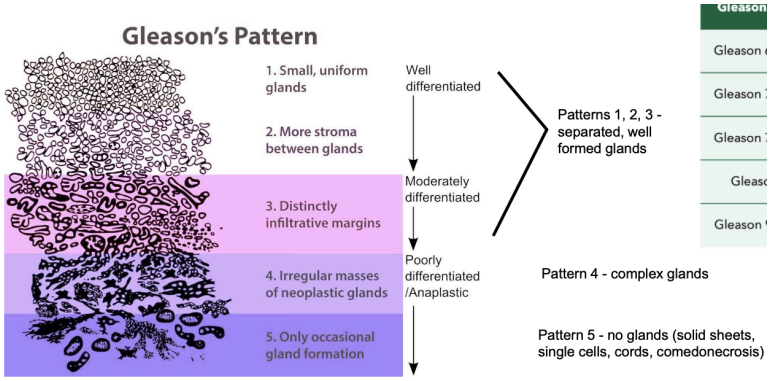


	 Benign hyperplastic glands lined by basal cells Serrated luminal border	
Stroma	Hyperplastic, **denser** stroma 	Tumor infiltrates the stroma 
IHC	Basal cell markers **positive** (brown); AMACR negative	Basal cell markers **negative**; **AMACR/p504s positive** (red)

		
What normal should look like by comparison		• The prostate is composed of glandular elements (known as acini) embedded in a fibromuscular stroma • Benign acini are lined by a peripheral layer of basal cells • Acinar cells are typical tall columnar cell with small, basally located nuclei • Nucleoli are inapparent 

Grading of cancer (Gleason): Here for reference, covered in other LOS

- Grading is based on **architecture, not cytology**. Patterns run from 1 to 5:
  - Patterns 3 and below: separate, well-formed glands.
  - Pattern 4: fused, **cribriform**, poorly formed glands.
  - Pattern 5: no glands; solid sheets, cords, single cells, or comedonecrosis.
- The **score is primary + secondary pattern**. **4+3 is worse than 3+4.**
- Rough interpretation: a score of 6 or less is least aggressive, 7 is intermediate, and 8 or higher is most aggressive.
- There is no grading for BPH because it is not a neoplasm.



5. Potential complications

- **BPH** complications all stem from **bladder outlet obstruction**:
  - **Lower urinary tract symptoms**: weak stream, hesitancy, nocturia, incomplete emptying.
  - **Bladder changes**: wall hypertrophy, **<u>trabeculation, and diverticula</u>**.
  - **Urinary retention and stasis**, which lead to **recurrent UTIs and bladder stones**.

    - **Hydronephrosis**, which can progress to **chronic kidney disease** and then **pyelonephritis/urosepsis**.
    - **Hematuria**.
    - **PSA elevation**: BPH is the **most common noncancerous cause** of a raised PSA.
    - **No malignant potential.** BPH and cancer can coexist, but one does not cause the other.
- **Adenocarcinoma** complications come from **local invasion and metastasis**:
    - **Local extension** (pathologic staging):
        - Extraprostatic extension or bladder neck invasion is **pT3a**.
        - **Seminal vesicle invasion is pT3b**.
        - Invasion of the rectum, levator muscles, or pelvic wall is **pT4**.
    - **Lymph node spread**: the first landing sites are the **obturator and external iliac** nodes.
    - **Bone metastases** to the **axial skeleton**. These are classically **osteoblastic** (sclerotic), causing bone pain and pathologic fractures; the spine is the classic site.
    - **Late obstruction** can occur if the tumor is large or invades the bladder neck. This is less common early than in BPH.
    - **Treatment-related harms of ADT**: hot flashes, loss of libido and erectile dysfunction, osteoporosis and fractures, metabolic syndrome, anemia, cardiovascular morbidity, and cognitive decline.

Recall the pathophysiology of prostatitis. PATH -**Tomiwa**

Prostatis:

- Inflammation of prostate gland caused by bacterial infection or noninfectious mechanisms

Acute Bacterial prostatitis:

- Bacteria trigger inflammatory response after entering prostate
- Inflammation causes prostate swelling and tenderness -> pelvic pain and urinary symptoms seen
- May progress to prostatic abscess., acute urinary retention or sepsis

Findings: fever, chills, perineal pain, dysuria, urinary frequency, urgency. Prostate will be tender and boggy

Chronic Bacterial Prostatitis:

- Persistent or recurrent bacterial infection that causes ongoing inflammation of the prostate
- May result from inadequately treated ABP or poor antibiotic penetration
- A lot of causative organisms will produce biofilm to shield from abx
- Tissue changes seen include **patchy, fibrotic areas of chronic inflammation** that will reduce antibiotic diffusion instead of diffuse inflammation seen in ABP
- Pts experience UTI and urinary sx eg dysuria, frequency, urgency

Chronic prostatitis/ chronic pelvic pain syndrome (CPPS)

- No evidence of UTI and exact cause is unknown.
- It is best understood as a multifactorial cascade that involves an event that triggers inflammation and immune activation that will ultimately lead to central sensitization and chronic pain. May also see erectile dysfunction
- Inflammation is inconsistent with some people not having any inflammation

Asymptomatic Inflammatory prostatitis (I believe this is low yield but for completeness)

- Presence of histologic or seminal inflammation with no clinical sx
- Usually an accidental finding

REcall:

- Urinary tract infections (UTIs) are infections of the bladder, urethra, ureters, or kidneys that are most commonly caused by bacteria, especially *E. coli*. Infections of the bladder or urethra are called lower UTIs, whereas infections involving the kidneys or ureters are called upper UTIs.
- Urinary tract infections (UTIs) and prostatitis are closely related because bacteria from a UTI can ascend through the urethra and infect the prostate gland, or a persistent prostate infection can act as a reservoir that causes recurrent UTIs

Bacterial Prostatitis:
Distinguishing Acute vs. Chronic Care

Bacterial prostatitis is categorized into Acute (ABP) and Chronic (CBP) forms, each requiring distinct diagnostic approaches and treatment durations. While *E. coli* is a common driver for both, the clinical presentation and management vary significantly.

ACUTE BACTERIAL PROSTATITIS (ABP)

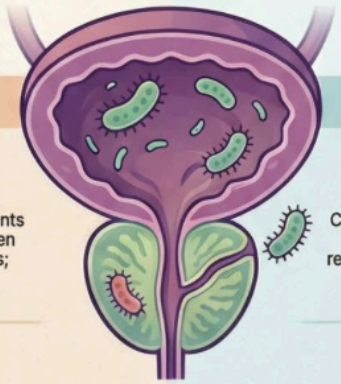
CLINICAL PRESENTATION & DIAGNOSIS

Sudden Systemic Illness

ABP presents with sudden fever/chills;

KEY FINDING:
Digital Rectal Exam (DRE)

Prostatic tenderness on brief, genfle DRE is a primary diagnostic indicator for ABP.



CHRONIC BACTERIAL PROSTATITIS (CBP)

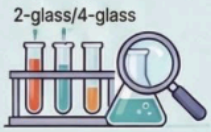
CLINICAL PRESENTATION & DIAGNOSIS

Indolent Recurrence

CBP is gradual with recurrent UTIs.

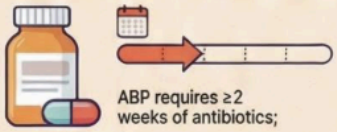
DEFINITION:
Meares-Stamey Test

The 2-glass or 4-glass test is the gold standard for confirming CBP.

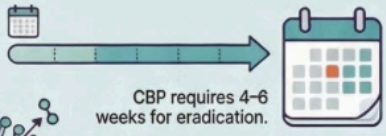


MANAGEMENT & TREATMENT PROTOCOLS

STATISTIC: **Divergent Antibiotic Durations**



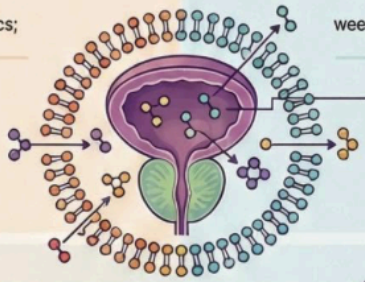
ABP requires ≥2 weeks of antibiotics;



CBP requires 4–6 weeks for eradication.

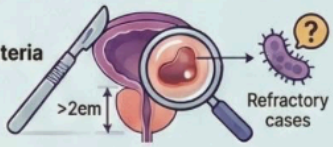
SUPPORTING FACT:
Overcoming the Prostate Barrier

Lipid-soluble drugs like fluoroquinolones are preferred for superior prostatic tissue penetration.

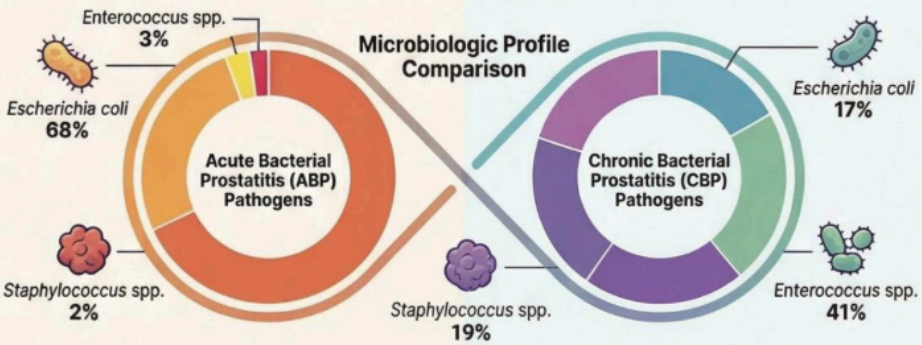


PROCESS STEP: **Surgical Intervention Criteria**

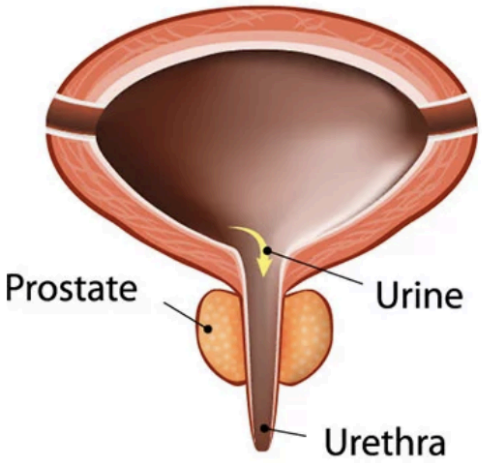
Reserved for prostatic abscesses >2cm or cases refractory to proionged antimicrobial therapy.



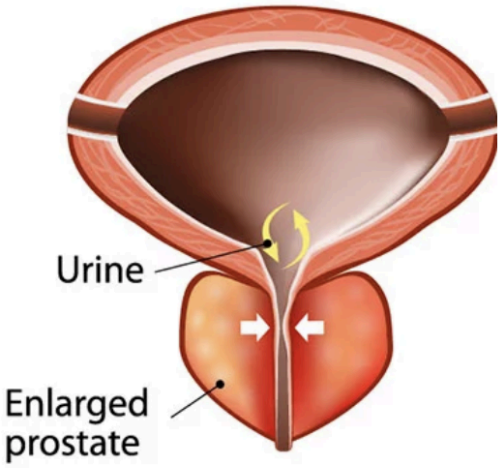
Microbiologic Profile Comparison



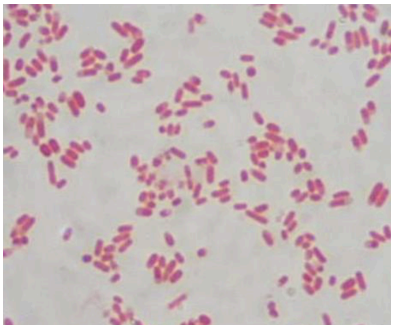
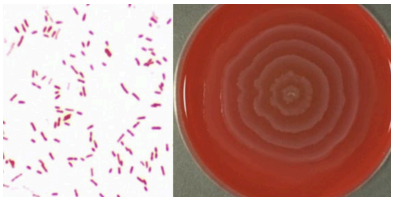
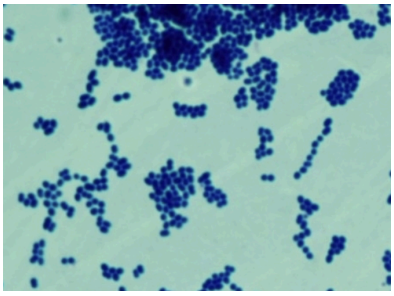
Normal prostate

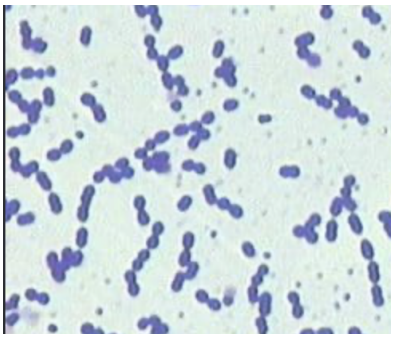
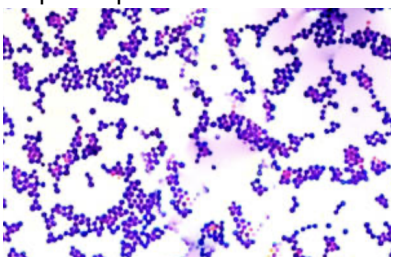
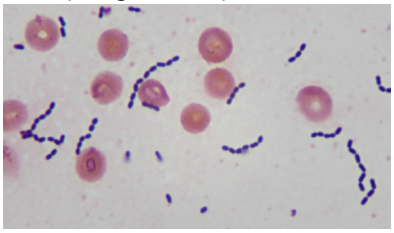


Prostatitis



8. Recall the common bacterial causes of urinary tract infection and their microbiology and virulence factors. INIM- **Tomiwa**

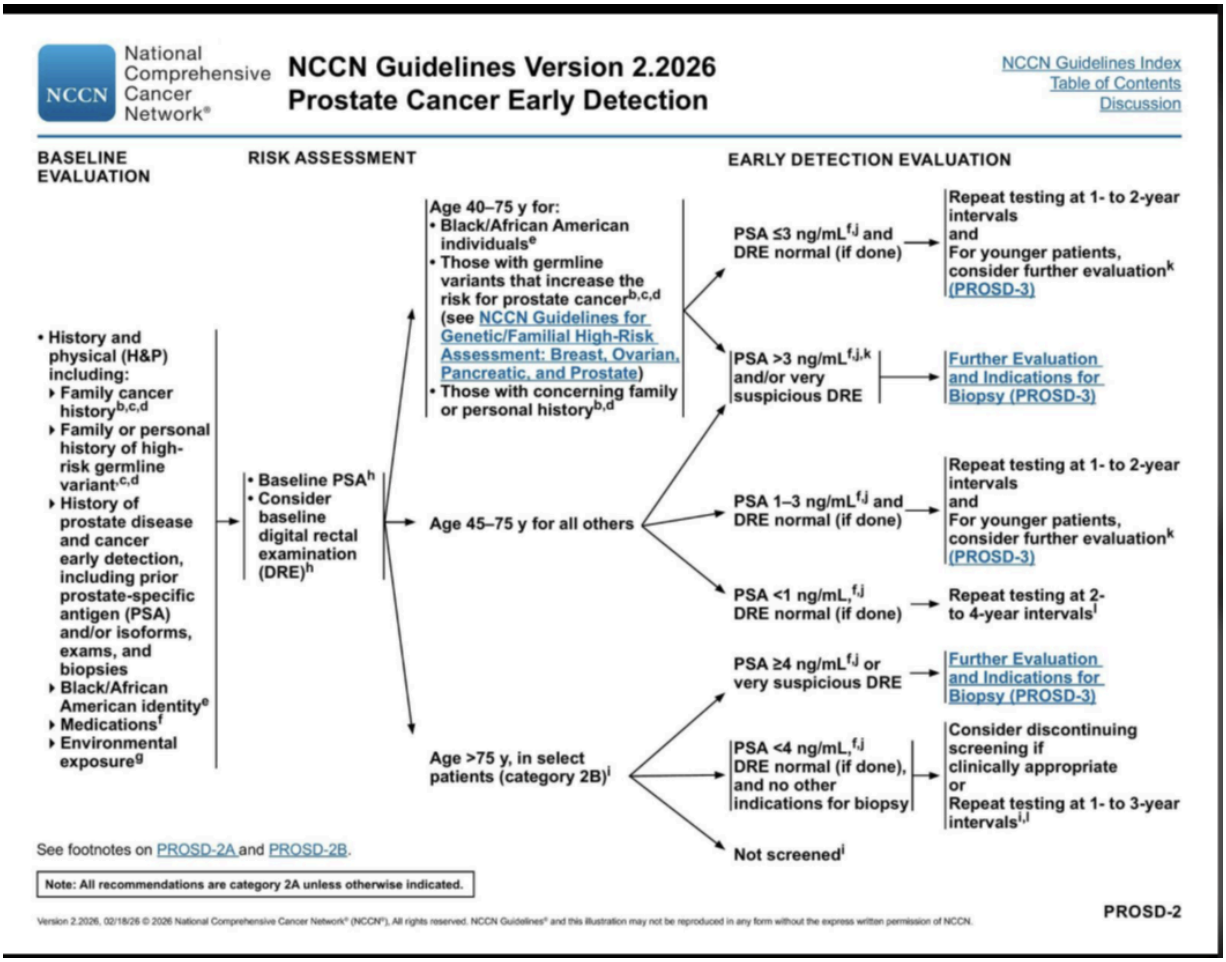
Bug	Microbiology	VF
E.coli 	- Most common cause of UTIs - G-ve rod - Facultative anaerobe - Lactose fermenter	Adhesins (fimbriae) promote attachment to urinary epithelium and help bacteria resist being flushed out by urine. The most common UTI pathogen
K. pneumo	- 2nd mst common - Gram-negative rod; - encapsulated; - lactose fermenter	Polysaccharide capsule helps resist phagocytosis; associated with UTIs, including healthcare-associated infections
Proteus mirabilis 	- G -ve - Motile rod - Urease +ve	Urease converts urea to ammonia, increasing urine pH and promoting struvite stone formation.
Pseudomonas aeruginosa	Gram-negative rod; non-lactose fermenter; oxidase-positive	Biofilm formation and intrinsic antimicrobial resistance contribute to healthcare-associated and catheter-associated infections.
Enterococcus Faecalis:  Faecium:	Gram-positive cocci, often in pairs or chains; facultative anaerobes	Can cause healthcare-associated and catheter-associated UTIs; notable for antimicrobial resistance, including vancomycin resistance in some strains.

		
Staph. Sapro. 	- G- +ve cocci in clusters - Coagulase -ve - Novoniocin resistant	Urease production; common cause of UTIs in sexually active women.
GBS (s. agalactiae) 	- G +ve cocci in chains - Catalase -ve - Beta hemolytic - CAMP-positive, hippurate-positive, PYR-negative -	Polysaccharide capsule helps inhibit phagocytosis
Enterobacter and serratia	Gram-negative rods	Opportunistic, often healthcare-associated pathogens; antimicrobial resistance is an important concern.

EPIDEMIOLOGY (incidence, prevalence, risk factors)

1. List risk factors for the development of prostate cancer and state how the presence of these factors may affect how you would counsel a patient on prostate cancer screening. CM-Tomiwa

Baseline risk assessment (NCCN PROSD-2) — for average-risk individuals, baseline PSA (and consideration of baseline DRE) begins at age 45–75 with a life expectancy ≥10 years. The following are moved to the age 40–75 higher-risk pathway.	
RFs: ● Advanced age (> 50 years)	Discuss PSA screening based on age, life expectancy and pt preference,
● Family history especially FDR	Explain increased risk and consider earlier screening - suspicious family history should prompt genetic testing - Begin discussion earlier (often 40–45)
● African-American descent : assoc w earlier onset and more aggressive clinical course	Discuss earlier screening (approx. 40) due to higher risk of more aggressive disease - consider annual screening; - personalize given higher overdiagnosis risk at older ages; - no benefit beyond ~75
● Genetic disposition (e.g., *BRCA2*, Lynch syndrome) : risk of aggressive ca wt a younger age	Discuss inherited risk and consider screening from a younger age - Recommend systematic annual PSA from age 40
● Dietary factors: high intake of saturated fat, well-done meats, and calcium	Discuss their association with risk, but do not use diet alone to determine screening eligibility.
Environmental exposure — Agent Orange exposure (e.g., many Vietnam War veterans) may be associated with increased risk of high-grade prostate cancer and should be disclosed in counseling	Disclose association with high-grade disease; factor into decision to screen



National Comprehensive Cancer Network (NCCN)

1. Compare and contrast the history and physical exam findings in patients with prostate conditions, including prostatitis, benign prostatic hyperplasia, and prostate cancer. CS

Prostatitis, BPH, and Prostate Cancer: History and Physical Exam
History

	Acute bacterial prostatitis	**Chronic bacterial prostatitis / CPPS**	**BPH**	**Prostate cancer**
Onset	Abrupt	Recurrent or persistent; CPPS is chronic	Gradual, progressive	Insidious; often found by screening (an elevated or rising PSA)
Systemic symptoms	High or spiking fever, chills, malaise	Usually absent; chronic bacterial prostatitis may cause a low-grade fever	Absent	Late: weight loss, fatigue
Pain	Severe perineal, pelvic, or low-back pain; pain with defecation	Mild pelvic pain in chronic bacterial prostatitis. In CPPS, chronic diffuse pain in the lower abdomen, pelvis, perineum, penis, scrotum, or low back, with no infection	Usually none	Painless locally. Late: bone pain, especially in the lumbosacral spine; flank pain from hydronephrosis
Urinary symptoms	Prominent irritative symptoms: dysuria, frequency, urgency, cloudy urine	Irritative: dysuria, frequency, urgency. Chronic bacterial prostatitis is suggested by recurrent UTIs with the same organism; CPPS by bladder-irritation symptoms	Obstructive (voiding): hesitancy, weak or intermittent stream, straining, terminal dribbling, incomplete emptying. Storage (irritative): nocturia, frequency, urgency. Sometimes hematuria	Often none early, because peripheral-zone tumors sit away from the urethra. Late: LUTS, retention, hematuria, incontinence
Sexual symptoms	—	Erectile dysfunction, hematospermia; painful ejaculation in CPPS	—	—
Zone of origin	Diffuse inflammation	Diffuse	Transition zone (periurethral, middle and lateral lobes). This is why LUTS start early	Peripheral zone (posterior and inferolateral). This is why the tumor is palpable on DRE but causes no early LUTS

Complications	Acute urinary retention, sepsis, prostatic abscess	Recurrent UTI; chronic pain and impaired quality of life (CPPS)	Acute retention, hematuria, recurrent UTI and urosepsis, bladder stones, hydronephrosis, CKD	Local invasion (seminal vesicles, bladder), lymphatic spread, osteoblastic bone metastases (axial skeleton), vertebral pain, spinal cord compression with neurologic deficits, lymphedema
Risk Ractor	UTI risk factors	Recurrent UTIs. CPPS is typical in younger to middle-aged men	Older age: about 50% of men in their 50s and 80–90% in their 70s–80s	Age: 80% of cases are over 65; 1 in 41 at ages 40–59 and 1 in 8 at 70 and older. Race: highest in Black and Scandinavian men, lowest in Asian men. Family history: first-degree relatives with prostate cancer (2–3 relatives raise risk) or with breast cancer. Western diet: risk rises in Asian immigrants as diet becomes "westernized." Androgens play a permissive role

Physical exam (DRE)

Tenderness on the digital rectal exam (DRE) is the main thing that separates these conditions:

	Prostate on DRE	**Exam pearls**
Acute bacterial prostatitis	Warm, swollen, **boggy, markedly tender** (**+ fever**)	Examine gently. Vigorous massage can cause bacteremia. Also look for fever, tachycardia, suprapubic tenderness, and signs of sepsis.
Chronic bacterial prostatitis	Often normal; may be enlarged or mildly tender	The rest of the exam is usually unremarkable.
CPPS	Usually normal; occasionally mildly tender	The rest of the exam is usually unremarkable.
BPH	**Symmetrically enlarged**, **smooth**, firm, rubbery (like the tip of the nose), **nontender,** with no discrete nodules	Check for a palpable or percussible (distended) bladder, which suggests retention. Look for neurologic causes of the urinary symptoms. Prostate size correlates poorly with symptoms.

Prostate cancer	Often normal, especially with early or anterior tumors. Suspicious findings: **a hard,** stony (knuckle-like) **irregular, nontender nodule**, induration, **asymmetry**, prostatomegaly, or loss of the median sulcus	DRE is the least sensitive way to detect cancer, and a cancer found because of a positive DRE is usually more advanced. Clinical stage cT1 is not palpable (cT1c is detected by PSA alone); cT2 is palpable but confined to the prostate; cT3 or cT4 has extended beyond the capsule. Also look for spinal tenderness, focal neurologic deficits, lymphadenopathy, leg edema, and cachexia.

Comparison points

- PSA can rise with any of the three conditions, and also with retention, urethral or prostatic trauma, an aggressive DRE, and, rarely, sex or ejaculation.
    - Elevated PSA alone doesn't tell you which condition is present.
- BPH is the most common noncancerous cause of an elevated PSA.
- Several things lower PSA:
    - Antibiotics, in about 30% of cases when infection is the cause.
    - **Finasteride or dutasteride, by about 50%**.(***50% keeps coming up)***
    - Statins, by about 4%.
    - Simply repeating the test.
- No PSA level is low enough to rule out cancer.
- BPH and advanced cancer can both cause LUTS, retention, and hematuria, so symptoms alone can't distinguish them.
    - LUTS = Lower Urinary Tract Symptoms; umbrella term for issues affecting the bladder, urethra, and the prostate in men. Includes…
        - Storage Symptoms: Trouble holding urine, like urgency, high frequency, waking at night (nocturia), and leakage.
        - Voiding Symptoms: Trouble getting urine out, like a weak or slow stream, starting and stopping, hesitation, and straining

Diagnostic:

- Neither PSA nor DRE can diagnose cancer; they only raise suspicion.
- Confirming cancer requires an ultrasound-guided biopsy, either transrectal or transperineal. A 12-core mapping biopsy is standard, with additional directed cores if imaging shows a lesion.
- Once cancer is diagnosed, PSA helps with staging: 0–10 is low risk, 10–20 intermediate, and over 20 high.

Classic cases (From TBL) ***Extra***

- **Case 2 (BPH):**
    - A 75-year-old with a weak stream, nocturia ×3, a sense of incomplete emptying, and a PSA of 4 that had been stable for years.
    - He developed a distended bladder, fever, and dysuria, followed by E. coli urosepsis and septic shock.
    - Autopsy: thickened bladder wall, enlarged nodular prostate, and a compressed prostatic urethra.
- **Case 3 (cancer):**
    - A 62-year-old with no symptoms whose PSA rose over 3 years.
    - DRE showed distinct firmness along the left side, and biopsy confirmed adenocarcinoma.

2. Describe the consequences of androgen undersecretion or deprivation in the postpubescent patient. CM- **Ayla P**

Androgen deficiency, or **hypoandrogenism**, can affect biological males and females. Testosterone has many roles in the human body, including:

- 46,XY: puberty, voice deepening, facial and body hair growth, sperm production, physical size
- Both sexes: skeletal muscle mass, bone density, erythropoiesis, metabolic energy, libido, sexual function

Hypoandrogenism can be **primary** (at the level of the testes/adrenals), **secondary** (at the level of the hypothalamus/pituitary), or a result of **androgen-deprivation therapy** for treatment of high-risk prostate cancer.

Effects of hypoandrogenism in adults include:

- 46,XY: decreased libido, erectile dysfunction, fatigue, loss of muscle mass, increased body fat, decreased testicle size, decreased facial or body hair growth, hot flashes, night sweats, decreased bone density, gynecomastia, depressed mood, anxiety, irritability, brain fog
    - Long-term use of ADT also increases risk of cardiovascular events and VTE
- 46,XX: decreased libido, sexual dysfunction, fatigue, decreased bone density, muscle loss
- **Symptoms are generally much more pronounced in males**

1. Recall the definition and application of the following features of diagnostic testing characteristics in the context of screening for prostate cancer: sensitivity, specificity, positive predictive value, negative predictive value. EBM

	Cancer on biopsy	**No cancer**
PSA positive	a = true positive	b = false positive
PSA negative	c = false negative	d = true negative

Term	**Definition**	**Formula**	**PSA screening (cutoff 4)**
Sensitivity	Men **with** cancer who test positive	a / (a + c)	**Low**, ~20%; many cancers missed
Specificity	Men **without** cancer who test negative	d / (b + d)	~90%; false positives from BPH, prostatitis, recent ejaculation, catheter, biopsy
Positive predictive value	Positive test that is truly cancer	a / (a + b)	~25% at PSA 4–10 (most biopsies negative); >50% at PSA >10
Negative predictive value	Negative test that is truly no cancer	d / (c + d)	~85%; PSA <4 does **not** rule out cancer

Pearl	
Prevalence	Sensitivity and specificity **do not** change. Higher prevalence (older, Black, family history) = **higher PPV, lower NPV**
Lower cutoff (4 to 2.5)	Sensitivity up, specificity down; more false positives and biopsies
SnNout / SpPin	Highly sensitive test, negative result rules out. Highly specific test, positive result rules in
Screening vs confirmation	Screening wants high sensitivity; confirmation (biopsy) wants high specificity
Problem with PSA	Low specificity and PPV: unnecessary biopsies, overdiagnosis of indolent cancer

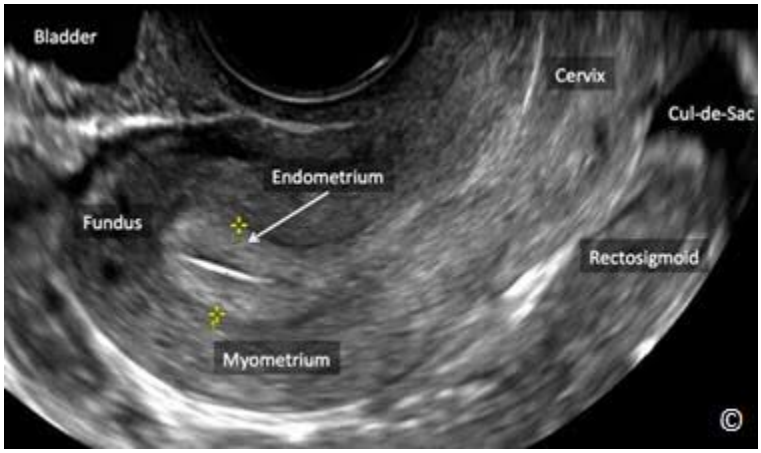
2. Identify the structures in the abdominopelvic region via imaging technologies. ANAT - **Ayla P**

The main structures visible in the pelvic cavity include:

- Organs: urinary bladder, ureters, urethra, sigmoid colon, rectum/anal canal, reproductive organs*
    - *46,XX: uterus, ovaries, fallopian tubes, vagina
    - *46,XY: prostate, seminal vesicles, testes
- Muscles: pelvic floor muscles, obturator internus, piriformis, psoas
- Bones: pelvic bones (ilium, ischium, pubis), sacrum, coccyx, femoral heads
- Vessels: iliac arteries/veins and branches
- Lymph nodes: external iliac, internal iliac, obturator, sacral

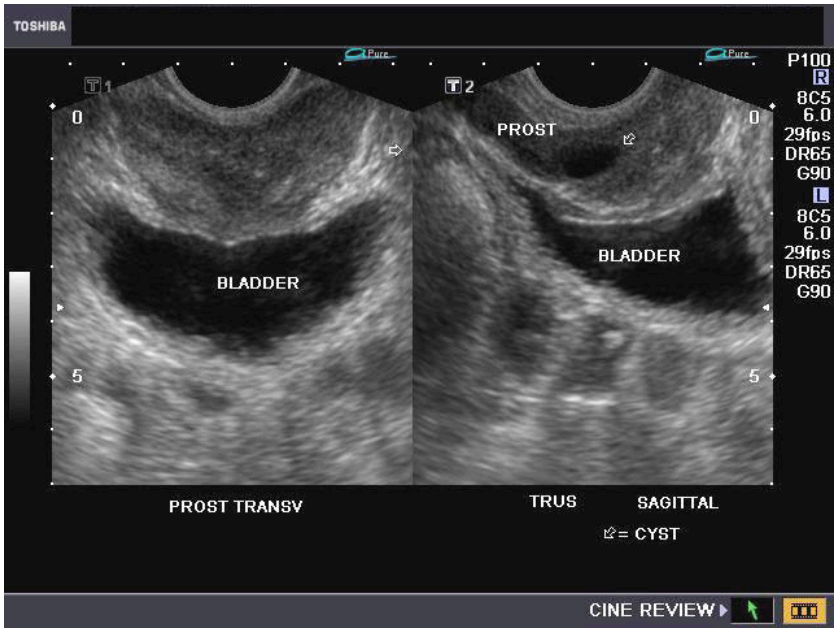
Imaging modalities include:

- Ultrasound
    - 46,XX: used for pregnancy monitoring, examination of the bladder, evaluation of masses, verification of IUD placement, examination of urinary tract



^example of (non-pregnant) uterine ultrasound

    - 46,XY: used for evaluation of prostate, seminal vesicles, rectum, and bladder



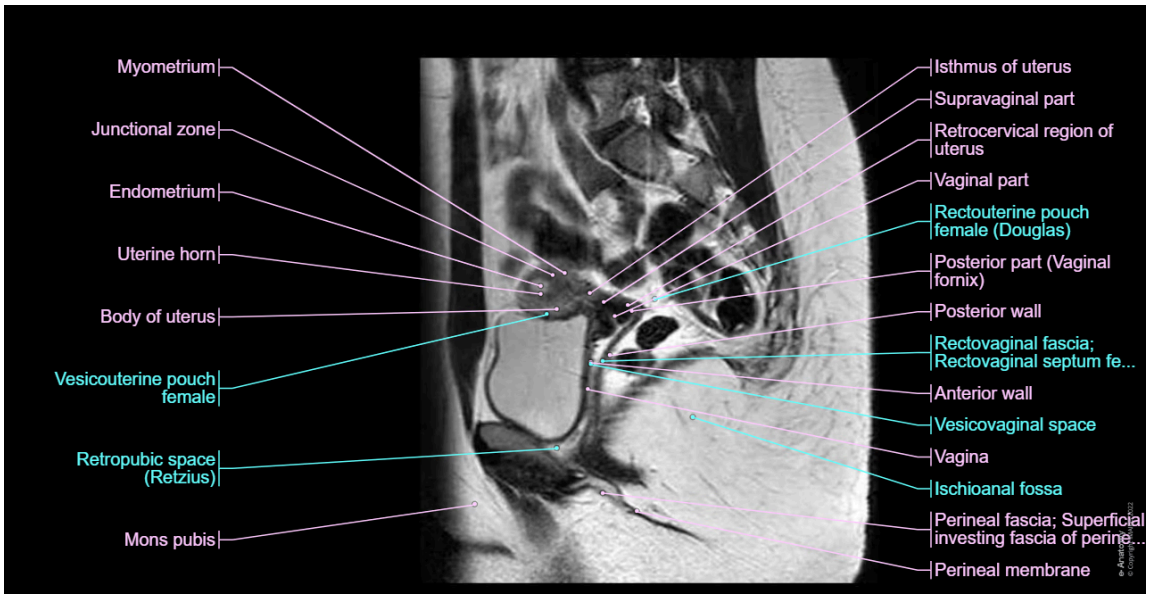
^example of prostate ultrasound

- CT: used to evaluate trauma, internal bleeding, acute pain, bony structures, joints, and kidney stones; also alternative to MRI in patients with metal

Link to female axial CT
Link to male axial CT

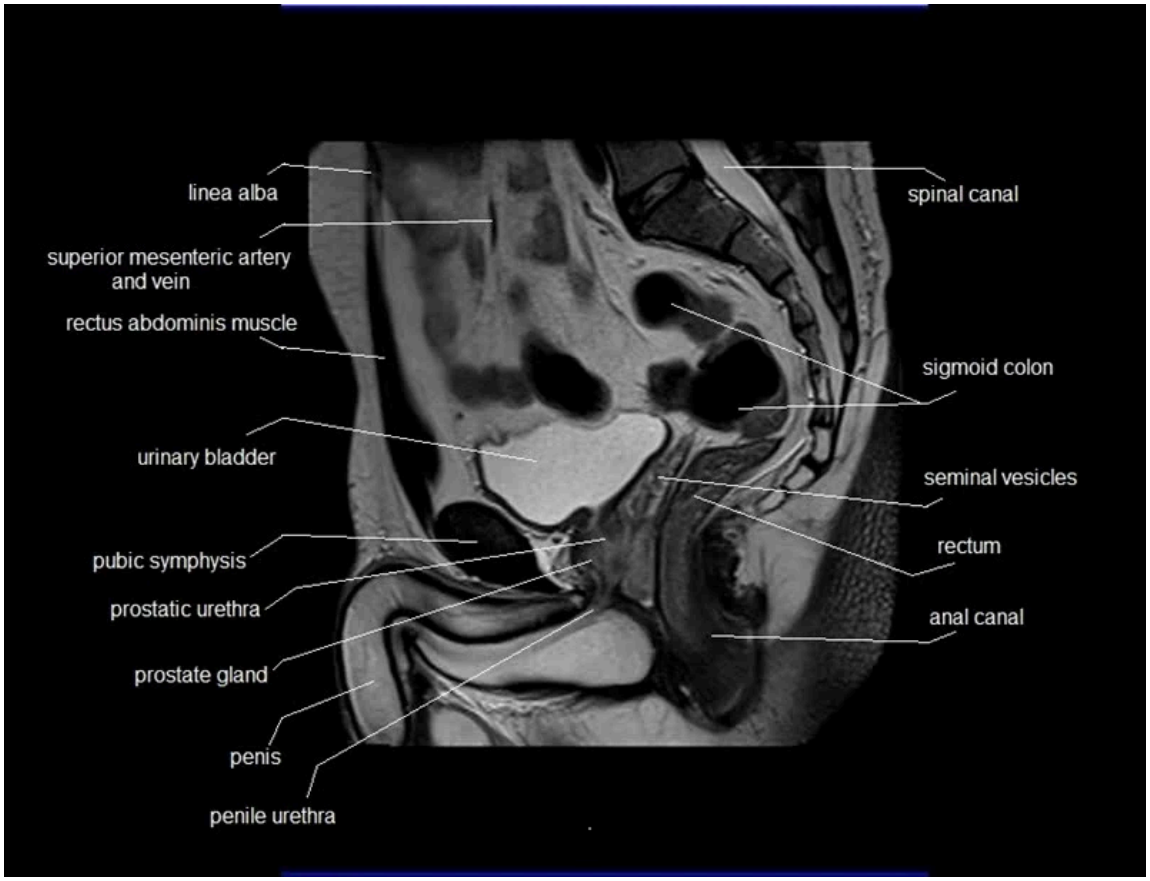
- MRI: used for soft-tissue detail of reproductive organs, muscles, and nerves; best option for unspecified pelvic masses

Sagittal view (female)
Link to full annotated scan



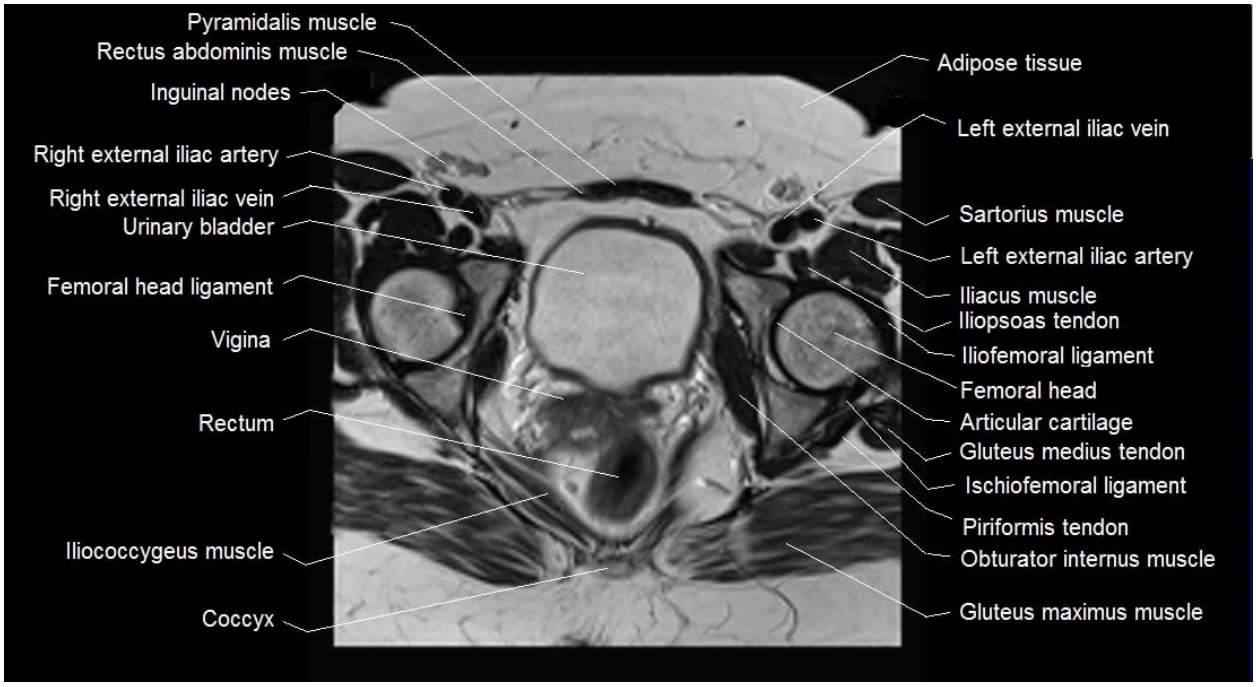
Sagittal view (male)

Link to full annotated scan



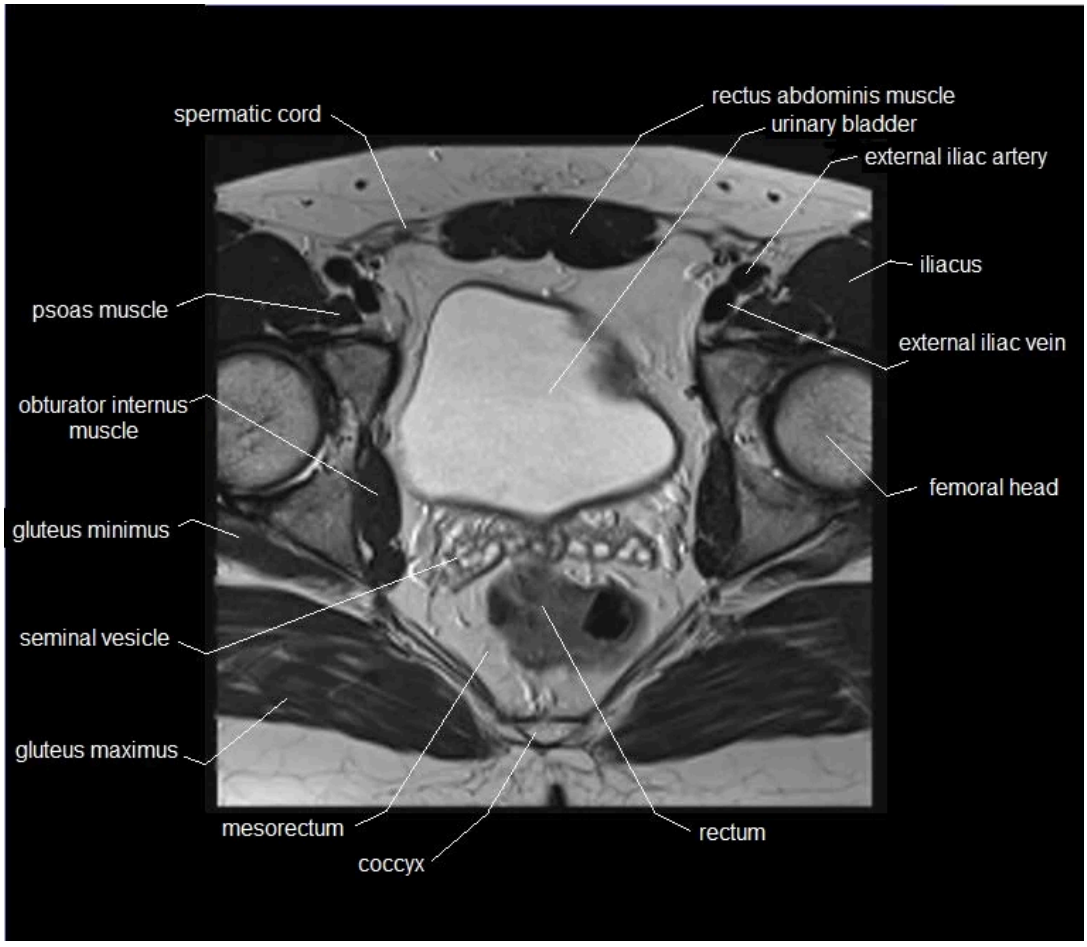
Axial view (female)

Link to full annotated scan

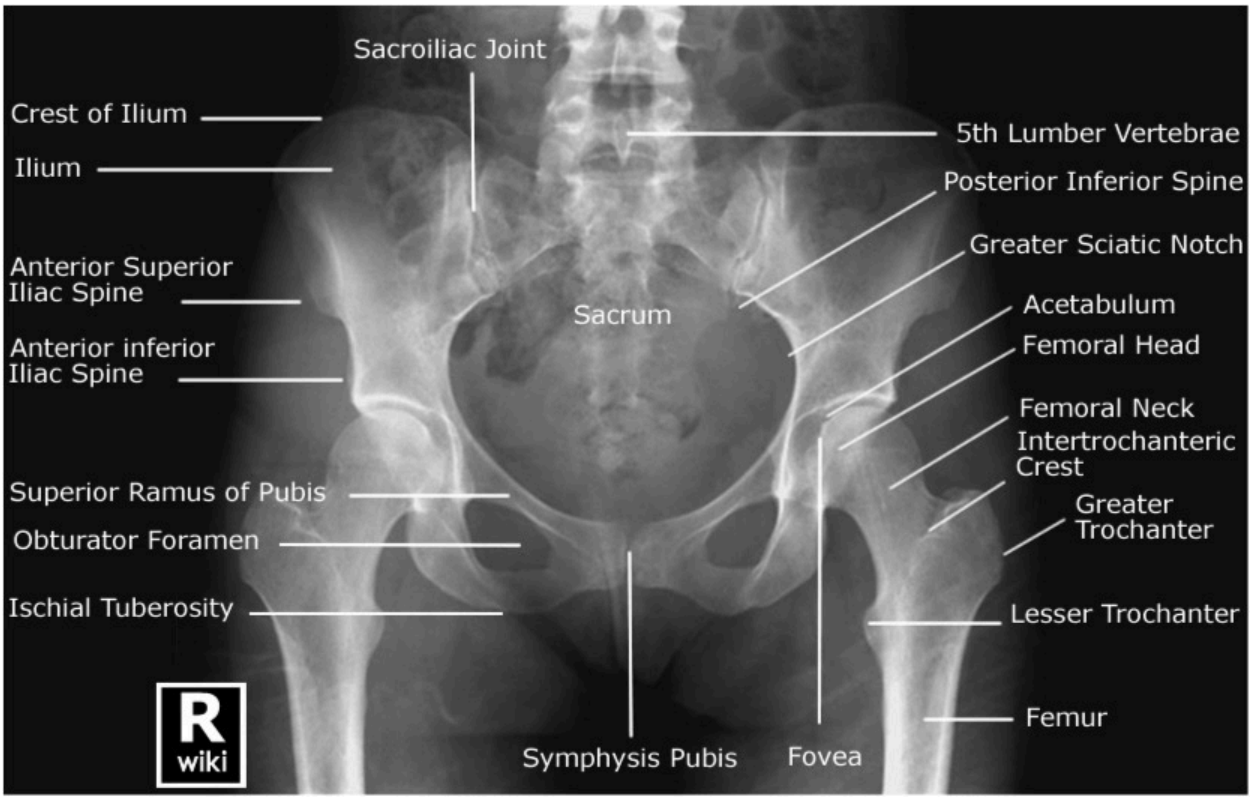


Axial view (male)

[Link to full annotated scan]



- X-ray: primarily used to evaluate bony structures for fracture, deformity, or misalignment



4. Discuss the differences between histopathologically detected prostate cancer and clinically significant prostate cancer, and explain how grade group, tumor volume, and other pathologic features determine which patients are appropriate candidates for active surveillance versus definitive treatment. PATH and CM

1. The core distinction

- **Histopathologically detected prostate cancer** means a pathologist has found adenocarcinoma in tissue. Says the cancer exists, not whether it will ever hurt the patient.
- **Clinically significant prostate cancer** is cancer with enough grade, volume, or anatomic extent to carry a real risk of progression, metastasis, or cancer death within the patient's lifetime.

The gap between the two is large:

- **Latent cancer is very common** → Men harboring prostate cancer rising steeply with age. By the 70s it reaches roughly 69–77%, far more than ever become ill from it.
- **The natural history varies → m**any cancers progress so slowly that the patient dies of something else first.
- **Screening finds indolent tumors.** PSA screening reduced prostate cancer deaths and metastatic disease at presentation. It also brought a real risk of **overdiagnosis and overtreatment**, with significant quality-of-life costs
- **Biopsy only samples the gland.** Most prostate cancers are multifocal. A 12-core biopsy can therefore under-sample the worst focus. Why **grade and volume on biopsy** are estimates, and why patients on surveillance need re-biopsy.

The clinical goal = Separate the two groups so that…

- Indolent tumors can be watched, sparing patients the incontinence, impotence, and infertility of treatment.
- Significant tumors are treated while they're still curable.

2. Establishing that it's cancer at all (histo not shown here, covered in previous LO)

Before deciding significance, the pathologist confirms malignancy.

- **Benign glands** have two cell layers: luminal secretory cells plus an outer **basal cell** layer. Lumens are often serrated or undulating, and corpora amylacea may be present. Nucleoli are inapparent.
- **Adenocarcinoma** shows:
    - small infiltrating glands with a **single cell layer** (no basal cells)
    - **prominent nucleoli**
    - intraluminal blue mucin or crystalloids
    - most often arising in the **posterolateral peripheral zone** → why it can be felt on DRE
- **Immunohistochemistry (CM):**
    - **AMACR (p504s)** is positive in cancer and HGPIN.
    - **Basal cell markers** are lost in cancer but retained in benign glands and HGPIN.
    - So "AMACR+, basal cell–negative" supports carcinoma.
- This step establishes *histologic* cancer only. Everything that follows determines *significance*.

3. Pathologic features that determine significance

A. Grade: Gleason score → Grade Group (the dominant factor) ***Covered in other LO here for reference***

Gleason patterns are based on architecture:

Pattern	Architecture	Meaning
3	Separate, well-formed individual glands with intervening stroma	Differentiated
4	Fused, poorly formed, or **cribriform** (sieve-like, back-to-back) glands	Loss of gland formation
5	No glands: solid sheets, cords, single cells, comedonecrosis	Undifferentiated

- The **Gleason score** is the sum of the most common (primary) pattern and the second most common (secondary) pattern.

Grade Groups regroup Gleason scores into prognostically distinct categories:

Grade	Gleason score	Clinical meaning
GG1	≤ 6 (3+3)	Indolent; rarely metastasizes. Mainstay of active surveillance.
GG2	3+4 = 7	Mostly pattern 3. Favorable intermediate; AS possible in selected patients.
GG3	4+3 = 7	Mostly pattern 4. Unfavorable intermediate; treat.
GG4	8	High risk
GG5	9–10	High risk

- **Teaching point: 3+4 and 4+3 are both "Gleason 7" but behave differently.**
  - The primary pattern drives behavior, which is exactly why Grade Groups separate GG2 from GG3.

B. Tumor volume (how much cancer) = Volume is estimated on biopsy using three measures:

- **Number of positive cores.** Very-low-risk disease has fewer than 3 positive cores out of a standard 12-core biopsy.
- **Percent involvement of any single core.** Very-low-risk disease has ≤ 50% involvement in every core.
- **PSA density (PSA ÷ prostate volume).** A value below 0.15 ng/mL/cc supports very-low-risk disease. A high PSA density suggests the PSA is coming from tumor rather than BPH.
- Note: On prostatectomy, the path report records the **percent of tissue involved by tumor**.

C. Anatomic extent (stage)

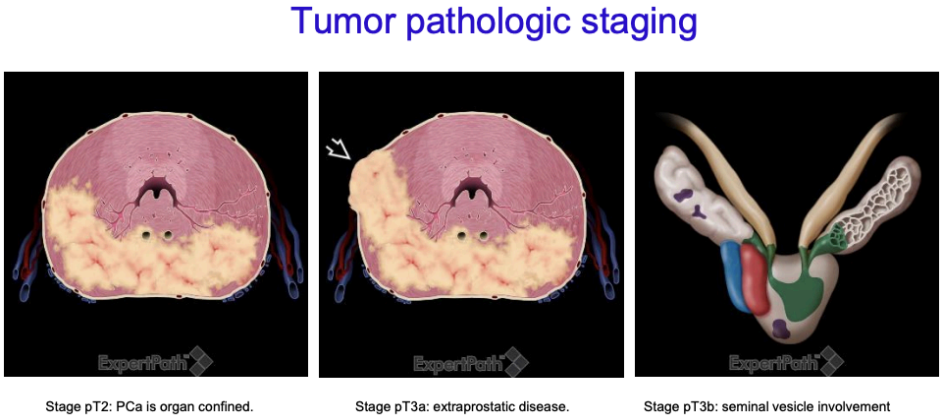
Clinical stage:

- **cT1c:** not palpable; detected only through PSA and biopsy.
- **cT2:** palpable but confined to the prostate.
- **cT3:** extracapsular extension, bladder neck invasion, or seminal vesicle invasion.
- **cT4:** invades adjacent organs.

Pathologic stage:

- There is **no pT1**, because you need the whole gland to stage.
- **pT2:** organ-confined.
- **pT3a:** extraprostatic extension or bladder neck invasion.

- **pT3b:** seminal vesicle invasion.
- **pT4:** rectum, levator muscles, or pelvic wall.



Margins:

- The surface of the prostatectomy specimen is painted with **ink**; Tumor touching ink is a **positive margin**.

Capsule:

- Tumor that stays inside the capsule is organ-confined. Tumor that crosses it is pT3 and has a higher risk of recurrence.

Note: **stage may carry more prognostic weight than grade alone**.

D. Other pathologic features

- **Perineural invasion** is common.
  - **Not** an independent prognostic factor.
  - Extensive perineural invasion can signal extension along nerves outside the capsule.
- **Cribriform architecture** is always pattern 4,
  - its presence moves a case out of GG1.
  - Many current guidelines also treat cribriform or intraductal carcinoma as an adverse feature that argues against active surveillance, even in GG2 disease.
- **Lymph node status** is reported on the prostatectomy specimen.
  - pN1 means the cancer is no longer localized.
- **PSA level** is not a pathologic feature, but it feeds into risk stratification:
  - < 10 ng/mL is favorable.
  - 10–20 ng/mL is intermediate.
  - 20 ng/mL is high risk.

4. Putting it together: risk group → management

Risk groups and management (CM lecture, clarified)

Risk group	Criteria	Management
Very low	cT1c, GG1, fewer than 3 positive cores, ≤ 50% in any core (PSA density < 0.15)	**Active surveillance**
Low	cT1c (or T1–T2a), GG1, PSA < 10, but not very low	**Active surveillance**

Favorable intermediate	Limited GG2, low volume, PSA < 20	AS in carefully selected patients; otherwise definitive therapy
Unfavorable intermediate	GG3, or multiple intermediate features, or high volume	**Definitive therapy**
High	GG4–5, PSA > 20, or cT3 or higher	**Definitive therapy**, often with ADT

Active surveillance (CM)

- AS is indicated primarily for very-low- and low-risk disease.
- Week 3 Notes 6 adds that it is reasonable in selected favorable-intermediate patients, provided **life expectancy is more than 5 years**.
- AS is **not doing nothing**. Monitoring includes:
    - serial PSA and DRE
    - **repeat biopsy every 6–12 months** at first
    - often multiparametric MRI
- Treatment is triggered by **upgrading** (for example, GG1 becoming GG2 or higher), increasing volume, rising PSA or PSA density, or new MRI or clinical findings.
- The patient may choose treatment at any time, even without progression.

Definitive treatment (CM)

Options are **radical prostatectomy** or **radiation** (IMRT or brachytherapy). The two have similar oncologic outcomes but different side effects.

- **Prostatectomy:** permanent infertility, usually transient stress incontinence, and possible impotence. Nerve-sparing is less feasible in higher-risk disease.
- **Radiation:** urgency and frequency, hematuria, rectal side effects, and possible impotence.
- **favored for:**
    - Intermediate risk (Gleason 7, i.e., GG2–3, or PSA 10–20), especially *unfavorable* intermediate.
        - Ex. T1c unfavorable intermediate-risk patient is **not** a surveillance candidate.
    - High risk: Gleason ≥8 (GG4–5) regardless of other favorable features, PSA ≥20, or cT2b–T4.
    - High-volume disease in the biopsy cores.
    - Extraprostatic extension (pT3a), seminal vesicle invasion (pT3b), invasion of adjacent organs (pT4), nodal disease, or metastases.
    - **Key point:**
        - A small Grade Group 1 tumor should not automatically trigger treatment.
        - A high-grade or high-volume tumor is clinically significant even while still confined to the prostate.

Watchful waiting

- This is different from active surveillance.
- It applies to patients with **life expectancy of 5 years or less**. There is **no curative intent**: symptoms are treated only if they arise, and there are no scheduled biopsies.

5. Describe the Gleason system of grading prostate cancer, including the histopathological diagnostic criteria for the diagnosis of prostatic adenocarcinoma, and how it relates to tumor biology. PATH

1. Diagnosing prostatic adenocarcinoma on histology

- First → know **benign** prostate looks like.
- Benign acini sit in fibromuscular stroma.
- Lined by tall columnar secretory cells with small, basally placed nuclei and no visible nucleoli. Underneath those cells is a **peripheral basal cell layer**, so benign glands are two cell layers thick.
- BPH glands keep this two-layer structure, often with a serrated or papillary luminal border and corpora amylacea.

Adenocarcinoma is diagnosed by combining architecture and cytology:

- **Architecture:** small, crowded glands that infiltrate between benign glands. They vary in size, have little stroma between them, and have straight, rigid luminal borders.
- **Loss of the basal cell layer:** malignant glands are lined by a single cell layer. This is the key diagnostic feature.
- **Cytology:** enlarged nuclei with **prominent nucleoli**.
- **Luminal contents:** lavender-blue mucin is often present in the gland lumens.
- **Invasion:** tumor grows into the stroma, and **perineural invasion** is common.
    - Perineural invasion is **not an independent prognostic factor**.
    - Extensive perineural invasion inside the gland, however, suggests the tumor may be spreading along nerves outside the capsule.
- **Immunohistochemistry (IHC) for difficult cases:**
    - **AMACR (p504s)** is positive (red) in cancer and in high-grade prostatic intraepithelial neoplasia (HGPIN).
    - **Basal cell markers** are negative in cancer but positive (brown) in benign glands and HGPIN.
    - So cancer is **AMACR+ and basal-marker−**. HGPIN is **AMACR+ but keeps its basal cells**.
- **Clinical and gross context:** most cancers arise in the **posterolateral peripheral zone**, which is why a digital rectal exam can feel a hard nodule. Tumors are often multifocal. On gross exam they appear as firm, poorly circumscribed, yellow-white or gray areas. Diagnosis requires an image-guided core biopsy.

2. The Gleason system

Gleason grading is based on **glandular architecture at low power**, not on how atypical the cells look. Each pattern is assigned a number from 1 to 5:

Pattern	Architecture
1–2	Well-formed, uniform, closely packed glands. These are rarely assigned in modern practice.
3	Discrete, separate, infiltrating glands of varying size, with stroma between them.
4	Poorly formed, **fused**, or **cribriform** (sieve-like, back-to-back) glands.

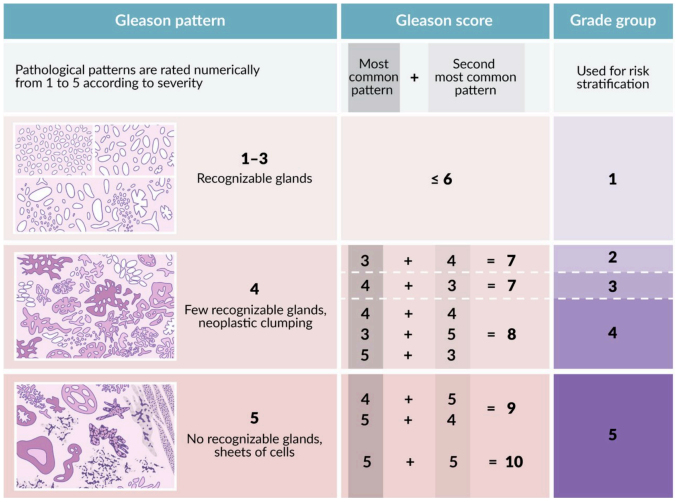
5	**No gland formation.** The tumor grows as solid sheets, cords, single cells, or nests, sometimes with **comedonecrosis**.

Gleason score = primary (most common) pattern + secondary (next most common) pattern.

- If only one pattern is present, it is doubled, so the possible range is 2–10.
- In practice, the lowest score reported on biopsy is 6 (3 + 3).
- The **order matters**. A 4 + 3 = 7 tumor is worse than a 3 + 4 = 7 tumor because pattern 4 makes up most of it.

Grade Groups convert the score into five prognostic tiers:

Grade Group	Gleason score
1	≤ 6
2	3 + 4 = 7
3	4 + 3 = 7
4	8
5	9–10



3. How grade relates to tumor biology

The Gleason pattern measures **loss of differentiation**: how much the tumor has lost the ability to form normal glands.

- Aggressiveness rises with score: ≤6 is least aggressive, 7 is intermediate, and ≥8 is most aggressive.
- Differentiation reflects behavior. Gland-forming tumors (pattern 3) tend to grow slowly and are often clinically insignificant. As tumors lose the ability to form glands (patterns 4–5), they grow more invasively and are more likely to spread outside the prostate, to lymph nodes, and to bone.
- **Low grade (Grade Group 1):** well-formed glands reflect a tumor that is still differentiated, growing slowly, and has low metastatic potential. These patients are often candidates for **active surveillance**.
- **Higher grades:** fused, cribriform, or solid growth reflects a tumor that is dedifferentiated, more proliferative, and more invasive. These tumors carry a greater risk of extraprostatic extension, seminal vesicle invasion, nodal spread, recurrence, and metastasis.
- Grade drives management. Grade group is a main input for risk stratification. GG1 can often be managed with active surveillance. GG4–5 is high risk and is treated definitively, even if other features look favorable.
- Grade works with other features. The radical prostatectomy report includes Gleason score, local spread (extraprostatic tissue, bladder neck, seminal vesicles, margins), lymph node status, and the percentage of tissue involved by tumor.
    - Together → determine prognosis.

Grade Group is described as the **most important prognostic factor** and reflects metastatic potential. It drives risk stratification together with:

- PSA level
- TNM stage
- Tumor volume (number of positive cores and % involvement of each core)
- Ex. very low risk is defined as Gleason ≤ 6 with T1c disease, fewer than 3 positive cores, and ≤ 50% tumor in any one core. High risk is Gleason ≥ 8.
- Summarized clinical gradient simply:
    - Score ≤ 6: least aggressive
    - Score 7: more aggressive
    - Score ≥ 8: most aggressive
- Stage still matters independently.
    - A radical prostatectomy report includes Gleason score, extraprostatic, bladder neck, and seminal vesicle extension, margin status, lymph node status, and percent of tissue involved.

1. Compare and contrast the MOA and side effects of alpha blockers and 5-alpha reductase inhibitors. CM and PHARM Ishaan

Pharmacologic treatment of BPH relies on relaxing smooth muscle to relieve immediate obstruction or shrinking prostate volume to address lower urinary tract symptoms (LUTS)

First line: Alpha Blockers
For men with enlarged prostates (typically 30-40 mL): 5-ARIs
Combination therapy is often recommended in men who have enlarged prostates and LUTS
Men with ED and BPH: PDE5

Drug Class	MOA	ADR
Alpha Blockers *1st line; rapid symptom relief* ● *Tamsulosin* ● *Doxazosin* ● *Alfuzosin* ● *Terazosin*	Blocks alpha-1 adrenergic receptor in the bladder neck and prostatic urethra → smooth muscle relaxation	Dizziness Orthostatic hypotension Headache Retrograde or premature ejaculation
5-Alpha Reductase Inhibitors (5-ARIs) *Indicated when prostate is enlarged* ● *Finasteride* ● *Dutasteride*	Inhibition of 5-alpha reductase → decreased conversion of testosterone to DHT → decreased prostatic growth + increased prostatic apoptosis and involution *Recall: testosterone is converted to DHT via 5-alpha reductase; DHT is the main hormone driving prostate growth*	Reduced libido Erectile dysfunction Decreased ejaculatory volume Potential mood changes

- 5-ARIs require 3-6 months of use for maximum prostate shrinkage
    - 5-ARIs are also used to treat androgenetic alopecia in males
- **5-ARIs cut serum PSA levels in half**
    - When calculating PSA to screen for prostate cancer, physicians should double the PSA for patients taking 5-ARI

1. Describe the principle of active surveillance for prostate cancer. CM

Active surveillance	**Key points**
Principle	Defer treatment of **low-risk, likely indolent** cancer; monitor closely and treat **with curative intent** only if it progresses. Avoids overtreatment (ED, incontinence, radiation proctitis).
Candidates	**Very low / low risk:** Gleason 6 (Grade Group 1), PSA <10, T1–T2a. Select favorable intermediate risk. Life expectancy **>10 yr**.
Monitoring	PSA ~q6 mo, DRE ~q12 mo, MRI, **repeat biopsy** (~1 yr, then periodically).
Triggers to treat	Gleason/Grade Group **upgrade** on biopsy, increased tumor volume, rising PSA, MRI progression, patient preference.
Then	Radical prostatectomy or radiation therapy.

	Active surveillance	**Watchful waiting**
Intent	Curative (delayed)	Palliative
Patient	Healthier, life expectancy >10 yr	Older/comorbid, limited life expectancy
Follow-up	Scheduled PSA, DRE, MRI, biopsy	No routine biopsy; symptom-driven
If progression	Surgery or radiation	Symptom control (e.g., ADT)

1. Describe the utility of PSA for surveillance/monitoring in patients who underwent or are undergoing prostate cancer treatment. CM

Treatment	Expected PSA	Recurrence
Radical prostatectomy	**Undetectable** by ~6 weeks	**≥0.2** (confirmed twice)
Radiation	Falls slowly, lowest point at ~2 years, never zero	**Lowest point + 2** (Phoenix)
Androgen deprivation therapy	Falls	Rising PSA with castrate testosterone = **castration-resistant**

Pearl	
Detectable PSA after prostatectomy	Residual or metastatic disease
PSA bounce	Temporary rise 1–3 years after radiation (esp. brachytherapy), **not** recurrence
Short PSA doubling time	Aggressive disease, higher risk of metastasis and death
Finasteride / dutasteride	Lower PSA by ~50%
Rising PSA	PSMA PET; after prostatectomy, salvage radiation

4. Demonstrate an understanding of the obligation of error disclosure (see reading in additional resource folder).
Note: I used ONLY the pdf reading attached to the resource document to answer this question, not the other 2 linked sources!

The obligation to disclose harmful medical errors is an established ethical mandate grounded in the fundamental right of patients and families to receive honest, compassionate information about their care. Medical professionalism requires clinicians to **put the patient's need for truth above personal or interprofessional anxieties**, such as fear of medical malpractice litigation, loss of referrals, embarrassment, or damage to collegial relationships. Because healthcare is delivered by multi-provider care teams, error disclosure is a **collective duty of professional self-regulation** rather than an isolated individual responsibility.

When an error involves another clinician, the obligation follows the principle to **"explore, do not ignore"** requiring clinicians to gather facts directly through peer discussions approached with curiosity rather than accusations before making statements to the patient. Where multiple clinicians are involved, **conducting a joint disclosure conversation is recommended** to maintain consistent messaging, avoid forcing the patient to piece together disparate facts, and demonstrate shared accountability. Ultimately, healthcare institutions carry the overarching responsibility to ensure transparent disclosure occurs (especially when communication breaks down or clinicians disagree!) by **cultivating a "just culture"** and providing institutional support such as disclosure coaches.

Definitions:

- **Harmful medical error:** adverse event resulting from medical care that causes harm to a patient, triggering ethical mandate for open and compassionate disclosure
- **“Explore, do not ignore”:** a clinician’s obligation to gather facts and engage directly in peer-to-peer discussions with an involved colleague before communicating with the patient, preventing speculative or inaccurate statements
- **Joint disclosure:** multiple involved clinicians participate together in communicating an error to the patient ensuring clear, unified messaging and demonstrating shared responsibility
- **Collective professional self-regulation:** evaluating quality and maintaining professional accountability is a shared responsibility across the medical profession rather than an isolated duty of individual clinicians
- **Just culture:** institutional environment of trust that encourages clinicians to report safety events while maintaining clear boundaries between acceptable behavior and unacceptable risk

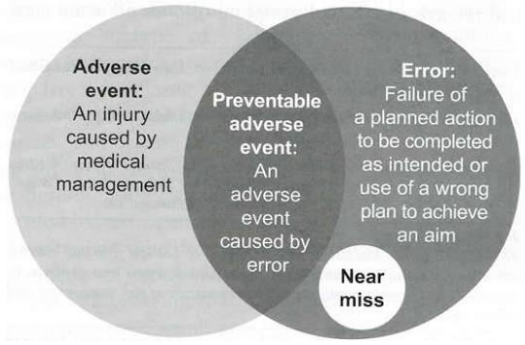
5. Review the classification of medical errors and the epidemiology of common errors. HSS. (Adverse Events, Near Misses, and Errors by the Agency for Healthcare Research and Quality)

Adverse events: harm from care instead of underlying disease

- **Preventable adverse events**: occurred due to error or failure to apply an accepted strategy for prevention
- **Nonpreventable adverse events**: injury resulting from medical management despite standard, non-negligent care
    - Ex: patient experiencing an embolic stroke during an indicated, standard coronary angiography despite not being at high baseline risk
- **Ameliorable adverse events**: events that, while not preventable, could have been less harmful if care had been different
    - Ex: heart failure patient discharged on furosemide without early lab follow-up instructions, develops acute kidney injury and critical hypokalemia 10 days later
- **Adverse events due to negligence**: occurred due to care that falls below the standards expected of clinicians in the community
    - Ex: delayed colon cancer diagnosis resulting from a physician performing only a limited sigmoidoscopy and reassuring a patient with persistent rectal bleeding

Hazards to patients: does not result in direct harm but is still unsafe

- **Near miss**: unsafe situation that is indistinguishable from a preventable adverse event except for the outcome
    - Ex: nurse inadvertently handing medication to the wrong patient in a shared room, where the mistake was caught by the patient before ingestion
- **Error**: a broader term referring to any act of commission (doing something wrong) or omission (failing to do the right thing) that exposes patients to a potentially hazardous situation



Epidemiology (how do we classify if an "adverse event" happened?):

- Two-stage record review process where patient charts are independently reviewed by two clinically experienced reviewers
- Even with highly trained reviewers, the level of agreement between reviewers with regard to the presence of an adverse event is not 100%
- As science of patient safety advances, these judgments can change over time, such that more adverse events become regarded as preventable

Describe the elements of full disclosure and apology when addressing those affected by medical errors. HSS. Reading: Chapter 18: The Malpractice System in Wachter, R. M., & Gupta, K. (2018).

Four steps to full communication after medical error = “TRAE”: Tell the patient/family what happened, Responsibility (you need to take it!), Apologize, Explain future prevention

Guidance to managing tricky cases with multiple parties:

Clinical Situation	Participants in Potential Disclosure	Rationale
Error involving a clinician at your institution who is, or was, treating a patient with you (e.g., a consulting specialist or colleague on a different service who previously cared for the patient)	Joint responsibility, with both clinicians participating in disclosure conversation	A joint discussion ensures that key information is communicated to the patient and demonstrates teamwork.
Error involving a trainee or interprofessional colleague (e.g., a nurse or pharmacist) on a primary team caring for the patient	Attending physician, with the person who made the error encouraged to participate in disclosure planning and the conversation itself (if appropriate)	The attending physician leads the care team and probably has the most experience with disclosure. Errors involving solely an interprofessional colleague could be disclosed jointly by the attending physician and the relevant manager.
Error involving a clinician at your institution who lacks direct contact with the patient (e.g., a radiologist or pathologist)	Attending physician on primary service treating the patient, with the colleague invited to join discussion	An existing patient–provider relationship facilitates disclosure conversations.
Error unrelated to current care (e.g., a radiologist reviewing a chest radiograph of patient admitted for pneumonia notices a retained foreign body from previous abdominal surgery)	Medical director (or other senior leader) at the institution currently caring for the patient, after consultation with clinician involved in error, with the current attending physician invited to join the discussion	The current treating clinician may not be well suited to explain an error unrelated to the present care. A senior medical leader is better positioned to handle this complex situation.
Error involving a clinician at another institution	Medical director at the institution currently caring for the patient, after consultation with the outside institution, with the current attending physician invited to join the discussion	The medical director can provide the patient with clinical information (on the cause and implications of the error) as well as administrative perspective. A local medical society or malpractice insurer may provide support for physicians who do not have access to institutional or organizational resources.

Disclosure program (from University of Illinois) describes seven pillars of an effective and comprehensive system of responding to adverse events:

“Pillar”	Explanation
1. Patient safety reporting system	Reports are “applauded”; can be made by phone, handwritten, online, or in person; clinical departments are penalized financially for failing to report incidents involving patient harm
2. Investigation	Risk manager performs preliminary investigation to ascertain harm. If yes, investigation and root cause analysis are commissioned. RCA to be completed within 72 hours of incident, including determination of whether “care was reasonable,” using Reason's algorithm of unsafe acts (Chapter 19)[4,28]
3. Communication and disclosure	The “centerpiece of the process”: ongoing communication with patient/family throughout process, including findings of “reasonableness” (#2). Third-party review offered if no consensus on whether care was reasonable. If “unreasonable,” then full disclosure and apology follow—but this is “a process” that plays out over time and avoids premature disclosure before clear conclusions. A “Patient Communications Consult Service” is available to aid in disclosure process; a member of the service is present at all disclosures
4. Apology and remediation	Combines saying “we are sorry” with offer of a remedy, with rapid remediation (including waiving hospital bills) and early offer of compensation, if warranted
5. System improvement	Even after apology, the investigation continues to identify and implement system improvements. Patients and families are invited to participate in these efforts
6. Data tracking and performance evaluation	Data regarding all aspects of this process are tracked and fed back into system for subsequent improvements
7. Education and training	Staff required to participate in initial and continuing training requirement on this process, including disclosure practices. In addition, providers are made aware of availability of a caregiver support program (“second victim”) for use after errors

8. Explain the mechanisms of action, adverse effects (major and common), and rationale for combined use of drug classes used as androgen deprivation therapy (chemical castration). PHARM Ishaan

Once a prostate malignancy has been staged, metastasis has been determined, and patient life expectancy has been accounted for, pharmacologic treatment (androgen deprivation therapy) may be indicated in

- Intermediate or high risk prostate cancer (high PSA and an unfavorable grade group)
- Locally advanced prostate cancer
- Metastatic prostate cancer

Androgen deprivation therapy (ADT) is often used in combination with radiation therapy, androgen synthesis inhibitors, and/or chemotherapy

Drug Class	MOA	ADR	Use
Androgen Deprivation Therapy	Decreased pituitary stimulation of androgen production in the testes - GnRH agonist (cause testosterone flair) - GnRH antagonist - GnRH receptor antagonist	- Osteoporosis/bone fracture - Sexual dysfunction - Gynecimastia, weight gain, decreased penis and testes size - Increased body fat, decreased muscle mass - Hot flashes and night sweats - Metabolic syndrome, cardiovascular toxicity, anemia	Use for locally advanced and metastatic prostate cancer as first line treatment Use as an alternative to radical prostatectomy for high-risk localized prostate cancer
Androgen Synthesis inhibitor	CYP17 gene inhibition → inhibition of androgen synthesis in adrenal glands, testis, and tumor tissue	- HTN - hypoKalemia - cardiac arrhythmias - Adrenal insufficiency	Adjunct therapy to ADT
Androgen Receptor Antagonist aka **Anti-Androgens**	Androgen receptor antagonist		Adjunct therapy to ADT Typically used for short-term management of a testosterone flare

9. Describe the mechanism of action, adverse effects, and clinically important drug interactions of PDE5 (phosphodiesterase 5) inhibitors in the treatment of erectile dysfunction. PHARM Ishaan

PDE5 inhibitors are first-line choices for symptomatic treatment of erectile dysfunction (often ingested orally)

PDE5 Inhibitor MOA, ADR, drug interactions

Phosphodiesterase type 5 inhibitor (PDE5 inhibitor)			
Names	**MOA**	**ADR**	**Drug-Interacitions**
Sildenafil Tadalafil Avanafil Vardenafil Dipyridamole	PDE5 inhibition → prevents cGMP breakdown → ↑ cGMP → enhances Nitric Oxide effect → ↑ smooth muscle relaxation pulmonary vasodilation, penile smooth muscle relaxation, and increased blood flow	**Generally well-tolerated** - Headache - Dyspepsia (relaxed LES) - Nasal congestion - Myalgia/back pain - Visual disturbance - Priapism ← *rare*	PDE5i's are hepatically metabolized **Contraindicated w/ Nitrates & nitrites** (nitroglycerin, nitrate "poppers") are contraindicated - because of the risk of hypotension, CV collapse and even death Use w/ alpha blockers may cause hypotension - Do not use within 2 hours of one another

- SildenaFIL, tadalaFIL, avanaFIL, and vardenaFIL: FIrst-Line therapy for FILling the corpus cavernosum
- PDE5 inhibitors work by amplifying the natural nitric oxide (NO) signaling pathway
- Indicated for ED, pulmonary arterial hypertension, benign prostatic hyperplasia (helps relieve LUTS by decreasing muscle tone)
- Contraindicated in patients taking nitrates due to the high risk of severe hypotension
- May cause orthostatic hypotension in patients taking alpha blockers